ASSA ABLOY

Control system Albany MCC



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These operating instructions are an integral part of the door. The original version of the operating instructions is in German.

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1 Introduction

These operating instructions apply to the use of the frequency converter control system MCC Vector Control with high speed doors. They are part of the product and describe the intended and safe use throughout the whole product life cycle.

Before the installation of the MCC read the installation and operating instructions carefully and familiarize yourself with their contents.

The manufacturer is not liable for damage caused by not observing of the operating instructions as well as their safety instructions, or a not intended use of the MCC.

1.1 Applicable documents

These operating instructions are only valid in combination with the operating instructions for the Albany high-speed door and the inspection book. Both of these documents are supplied with the product.

1.2 Declaration of conformity

The control unit is an integral part of the door. The declaration of conformity applies to the high-speed door as well as the control unit. It is included in the inspection book supplied with the door.

The control unit is designed in compliance with the requirements of the EN 13241-1 standard for door products and the subordinate standards mentioned in this standard.

Modifications that affect the technical data specified in the operating instructions or intended use, i.e. modifications that materially alter the machine, invalidate the manufacturer’s declaration of conformity.

1.3 Amendments and validity

The information contained in these instructions constitutes the technical specifications applicable at time of publication. Significant changes will be incorporated in a new edition of the operating instructions. The document and version number of these instructions can be found in the footer.

1.4 Document storage

Keep these operating instructions in the storage bag inside the control cabinet or in the supplied document box.

1.5 Target group

These operating instructions are addressed to trained, qualified and authorised personnel. All setup configurations, and in particular repairs and maintenance, may only be performed by employees of the manufacturer or other trained specialists.

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2 Safety instructions

Before installation, operation, service, or maintenance work on this high speed door or their electrical components it is mandatory to read and to comply with the safety instructions.

In addition to the general safety instructions in this section, always observe the safety instructions related to specific actions that are included in other sections. In particular, the general safety instructions in the operating instructions for the door are also applicable.

A nonobservance of these safety instructions can lead to damage of the door, injury or death.

2.1 General safety instructions

- Observe the general, legal and other national valid regulations for accident prevention and acknowledged technical rules for safety-oriented work.
- All work on electrical systems and connections may only be performed by qualified personnel with appropriate authorisation.
- The electrical work must be carried out according to the applicable local standards and legal regulations.
- Before performing any work on the control unit, cables, connectors or electrically live components, always switch off and lock the main switch or disconnect and secure the CEE plug.
- The door may be only operated, if the door is equipped with a disconnect device for the main supply voltage (such as a suitable main switch or a CEE plug). The main power disconnect device must be easily accessible and located between 0.6 and 1.7 meters above the access level.
- Make no change to the control or the safety devices which may affect the safety of the door.
- In order to avoid damages by electronic discharge, the circuit board of the control system and of the display should not be touched without appropriate ESD protective equipment.
- Some parameters are protected by an access code for safety reasons. The necessary access code level is given in square brackets after the chapter heading. Detailed information about the access code can be found in chapter 3.2.4.

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2.2 Safety instructions for operating the door

- The door is assembled in accordance with the installation instructions.
- All wiring between the door, the control unit and the drive is connected properly.
- All safety devices and protection equipment are installed as intended use and are functional.
- Safety devices are not modified or out of operation.
- The user must be instructed to leave the main power on at all times, to avoid the build-up of condensation in the control system housing.

2.3 Safety instructions for installation and maintenance

- Ensured that neither people nor vehicles pass through the door or are in the door hazard area during installation and maintenance.
- Dangerous touch voltage applies at the control up to 30 seconds after switching off/disconnecting the supply voltage (discharge of the capacitors). Therefore wait min. 30 seconds before working on electrical parts, wiring or opening the MCC. The non-observance can lead to an electric shock with serious injuries or death.
- A dangerous voltage at the MCC is applied after switching off/disconnecting the MCC power supply in case of doors with an uninterruptible power supply (UPS). The following instructions must be observed carefully:
  - ⇨ Switch off the main switch of the UPS as well.
  - ⇨ For working inside the UPS, the internal switch must be switched off (observe the operating instructions for the UPS).
  - ⇨ The non-observance can lead to an electric shock with serious injuries or death.
- Touching of electronic components on the MCC Board should generally be avoided.
- When opening the housing cover of the MCC the control system must be protected against moisture.

2.4 Safety instructions for “hold to run mode”

- In hold to run mode <u>all</u> safety devices are disabled. The door moves as long as the opening or closing button is pressed and the door has not reached the upper or lower end position. The door does not switch off until it reaches the upper or lower end position.
- Do not operate the door if any person or object is located in the door hazard area.
- Operate the door only if door area is visible.

3 Product Description

The MCC control system is designed to be used with high speed doors. The MCC meets the valid and applicable European CE and American/Canadian UL requirements, which has been tested and confirmed by independent testing institutes. The control system contains all necessary components to operate these doors with consideration of all safety criteria.

The characteristics of the MCC Control System with Vector Control allow even smoother starting and stopping of doors than is possible with standard frequency converters. This protects the door's mechanism against hard impacts and increases its speed.

3.1 MCC main unit with control box

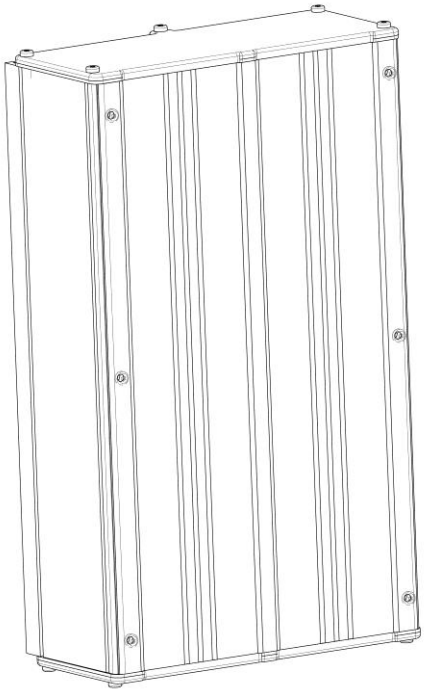


Figure 1: Control box MCC closed

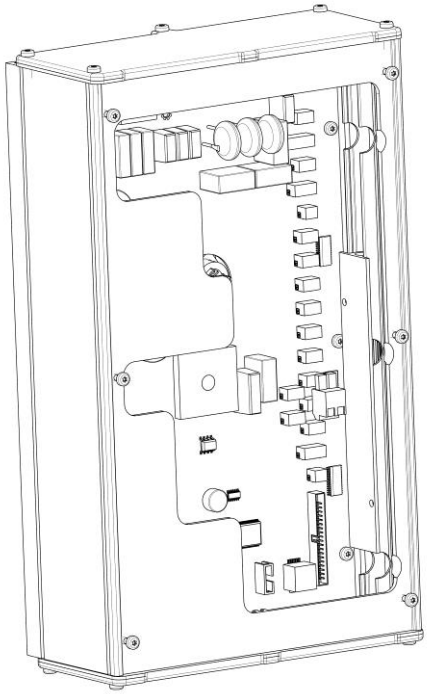


Figure 2: Control box MCC opened

The door control is built in an anodized aluminium housing. There is a connecting plate at each end for the introduction of the cables and for the junction box to be mounted.

Inside the housing (see Figure 2) the following components are located on the control board:

- The plug-in connection terminals for the inputs and outputs of the control system.
- The connection terminals for the net or primary supply power, motor and the brake cables
- The frequency converter including heat sink
- The short-circuit-proof power supply for the supply of the complete microprocessors, the frequency converter and external 24 V loads.
- Brake control

3.1.1 Safety devices of the control system

The closing sequence of an automatic door is monitored by various safety devices in order to avoid accidents. The safety devices and functions incorporated in the control unit are described below.

The doors are also equipped with other safety devices, depending on the door model. See the 'Safety Systems' section of the door user manual for information on the safety devices provided with the various models.

As electric safety devices, a pre-running safety photocell or an electrical safety edge can be used, which are complemented by a door line photocell in the side frame.

The electrical safety circuits of the MCC control system fulfil the requirements of the product standard for doors EN 13241-1.

3.1.1.1 Testing the safety devices

Before each closing movement of the door the fault free operation of the safety devices and the door line photocell are tested. If a fault is detected, no release of the closing movement occurs and an error message is displayed by the control. In case of an error the control changes in the hold to run mode, which allows the manual operation of the door.

Observe the safety instructions for “hold to run mode” (chapter 2.4).

3.1.1.2 Electrical safety edge (only for some types of door)

The electrical safety edge serves as closing edge protection. It consists of a rubber profile fitted to the main closing edge of the door. Any deformation of the rubber profile initiates an electrical impulse. This impulse immediately stops the door while it is closing and reverses regardless of the set opening height, up to the reference limit switch and is locked there. Locking is indicated by the following message on the display.



Pressing the “Open” or “Close” buttons releases the locking of the door.

However, the function can be adjusted such that the door automatically closes after a set time after the electrical safety edge has been activated (chapter 4.18).

Just before the lower end position of the door (max. 50 mm – 2”) the electrical safety edge is switched off because otherwise an activation via the ground would take place.

The MCC input for this fulfils the requirements of Category 2, Performance Level d of the EN ISO 13849-1.

3.1.1.3 Pre-running safety photocell (only for some door types)

The pre-running photocell runs approx. 100 mm (4”) below the bottom profile (depending on the door type). It moves with the door as it opens and closes. It retracts into the bottom profile when the door is in the lower end position. Just before the lower end position of the door (max. 50 mm – 2”) the prerunning photocell is switched off because otherwise an activation via the ground would take place.

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If the pre-running photocell is activated while the door is closing the door stops immediately and then opens again until it reaches the reference limit switch (reversing). After the tenth activation of the pre-running photocell (without the door closing completely) the door will be locked in the upper end position. It is however possible to parameterize the function differently (chapter 4.18).

The MCC input for this fulfils the requirements of Category 2, Performance Level d of the EN ISO 13849-1.

3.1.1.4 Door line photocell

The door line photocell is mounted in the lower area of the side frame of the door where it is protected. The proper function is monitored also by the control system. An activation of the photocell while the door is open resets and starts the hold-open time or the close after passage time. An activation of the photocell while the door is closing the door stops immediately and then opens again until it reaches the reference limit switch (reversing). Unlike for the electrical safety edge the door is not locked in this end position, instead it closes again automatically after the hold-open or close after passage time, if the door was closing in auto close mode while the photocell was activated. Otherwise it will stay open. As an option it's possible to get at some door types a light curtain instead of the door line photocell.

3.1.1.5 Emergency stop

The MCC control unit has provision for connecting two emergency stop devices corresponding to Stop Category 0 (stopping by immediate interruption of the power supply to the motor).

The MCC input for this fulfils the requirements of Category 3, Performance Level d of the EN ISO 13849-1.

3.1.2 Operating modes

The control unit has several operating modes, which can be selected by means of the menus. The following are the main operation modes.

3.1.2.1 Open by impulse / close after time

After an Open impulse, the door travels until it reaches the door open end position. The closing sequence starts after the hold-open time expires. The door closes until it reaches the lower end position. The hold open time can be configured during setting door limits (chapter 4.7.1 or 4.7.2) or separately (chapter 4.10).

If the door is closed, an Open impulse from the external impulse activator or the Open button on the display initiates the door to open immediately. If the door is open, the Open impulse restarts the hold-open time.

By a Stop command, either external or on the display, the door stops immediately and remains in this position until an operation of the door in the desired direction is initiated by an impulse activator. The display stop can be acknowledged only by an up or down to command on the display, the display stop is not acknowledged by external actuators.

If the door is in the fully opened state, a Stop signal terminates automatic closing and the door remains open until the next impulse. If this is followed by a Close impulse, the closing sequence starts immediately. An Open impulse restarts the hold-open time and initiates the door to close automatically after the time expires.

Interruption of the door line photocell or other safety devices the door reverse to the open position and closes after the hold-open time.

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3.1.2.2 Toggle circuit / manual closing

The toggle circuit / manual closing allows the door to be opened and closed using a single impulse activator. As long as the door is not fully open, every incoming impulse is interpreted as an Open impulse. An impulse received when the door is open starts a closing sequence. For safety reasons, only manual impulse activators (pull switches or buttons) should be used with this function. Program the input accordingly (chapter 4.14).

Interruption of the door line photocell or other safety devices the door reverses and remain in the open position. Closing of the door is started by a new impulse.

3.1.2.3 Close after passage function

If the close after passage function is activated, the door is opened by an Open impulse activator. The closing sequence is initiated by the interruption of the door line photocell and the door closes after the set hold-open time (Close after passage function). The door closes after the hold open time or after a close impulse if the door line photocell is not interrupted.

The settings for the close after passage function are described in chapter 4.13.

3.1.2.4 Hold to run mode

The door moves as long as the button is pressed and the door has not reached the upper or lower end position. The door switches off automatically when it reaches the upper or lower end position. All safety devices are disabled in hold to run mode.



Observe the safety instructions for “hold to run mode” (chapter 2.4).

3.1.2.5 Interlocking

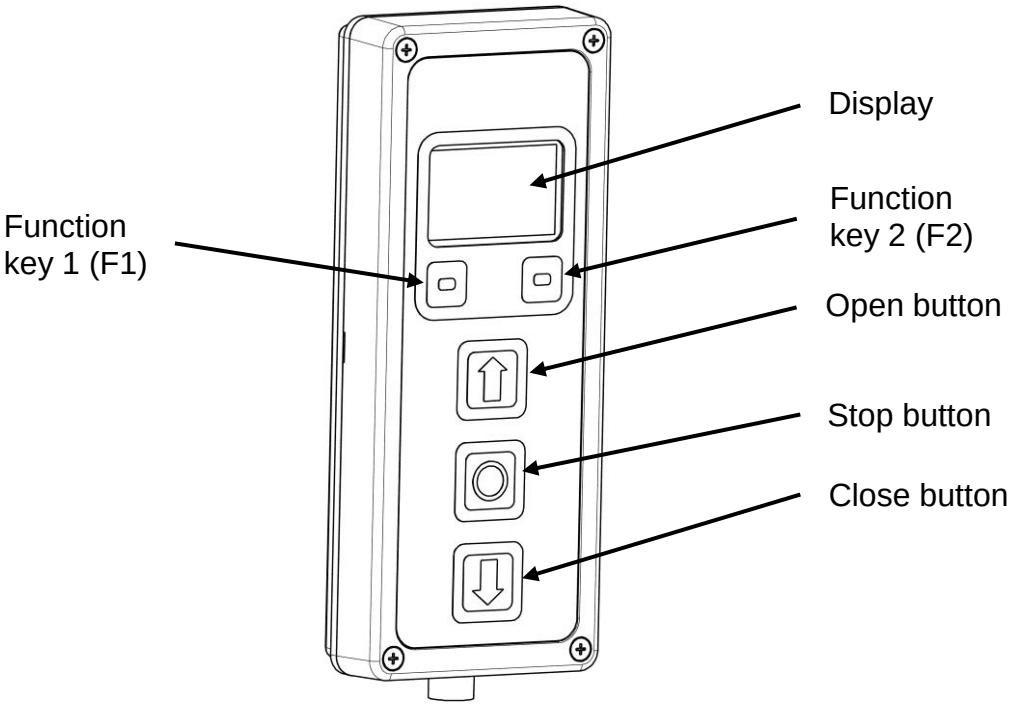
If two or more doors are interlocked, the control units are switched such that only one door can be opened at the same time. If one door is open or opening, all other doors are locked electrically and cannot be opened. The deactivation of the interlocking is described in chapter 5.11.

3.1.2.6 Airlock function

An airlock consists of 2 doors which are automatically controlled by the airlock sequence. Each door has a separate control system. An airlock is used, e.g. to prevent draughts in external doors. The installation of the airlock function is described in chapter 4.16.

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3.2 MCC - Display



The three buttons “Open (reset)”, “Stop” and “Close” are designed to control the door and to reset the MCC in the case of a malfunction. They can also be used to alter the individual settings (e.g. the hold-open time, etc.). The function of the two function keys is shown on the display at all times.

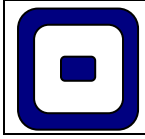
The display is not mandatory for operating the door. The general function can be realized through an external 3-pushbutton station.

3.2.1 Use of MCC - Display

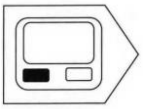
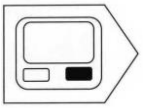
The display shows messages, faults, etc. in plain text or graphically. Faults during normal operation will be indicated immediately (chapter 6.7). For doors without a display the status light flashes (chapter 6.6).

	**“Open” button** This button works as an “open” button, sending an opening impulse to the control system. In addition, this button can be used to scroll upwards or to increase values in the individual menus.
	**“Stop” button** The door can be stopped at any point using the “Stop” button. Simultaneously the door is locked and can only be operated again once the “Open” or “Close” buttons on the display have been pressed. The stop function is shown on the display as plain text.
	**“Close” button** This button sends a closing impulse to the control system. Because the safeties are checked after every closing impulse there is a short delay before the closing sequence starts. In addition, this button can be used to scroll downwards or to decrease values in the individual menus.

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	**Function keys 1 and 2** The function of these two keys varies and is always shown on the display immediately above the keys.

The following symbols are used in the instructions to show that the keys should be pressed.

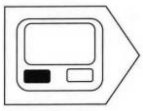
	Press left-hand function key!
	Press right-hand function key!
	Scroll up or down using the “open” or “close” keys

3.2.2 Save the input

The individual inputs are confirmed by pressing the button “Save”. If several digits are changed in a view, these must be saved separately.

3.2.3 Exit the input

The input or parameterisation of the MCC control system with the MCC display can be terminated by pressing (up to three times) the left function key “Escape”:



Alternatively, if no activity occurs, the display returns after a predefined time into the basic view.

3.2.4 Access codes MCC - Display

The MCC display enables the parameterisation of the MCC. Due to the ability to change safety-related parameters, the access to the individual parameters is limited by different levels of access codes. Hereby it is possible that there is no access to some parameters.

The necessary access code level is shown in brackets in the headers. There are following options:

No designation	=> accessible to all
[1]	=> customer code required (Factory setting 1111)
[2]	=> installation code required
[3]	=> system code required

4 Installation

Check the general operation of the door and the safety devices for proper function after installation of the door.

4.1 Mechanical installation of the control system

The control box should be mounted with the supplied mounting bracket. It can be also mounted onto an oscillation- and vibration-free wall.

CAUTION

Risk of material damage

Improper installation can cause damage to the wiring.

⇨ Ensure that the wiring is routed correctly during the control box installation.

⇨ When mounting the motor it must be ensured that the control system is not hanging on the cables.

Unused entry points should be sealed with appropriate sealed plugs. Any unused plug-in slots in the junction box must be closed with rubber stoppers. Only cable fittings with at least IP 55 rating may be used.

4.2 Electrical installation of the control system

4.2.1 Electrical installation requirements

For proper operation of the control unit, the electrical installation must fulfil the following requirements:

- The electrical work must be carried out according to the applicable local regulations and requirements.
- An external pre-fuse with 10A - 16A B- or C-characteristic or comparable must be planned locally for the MCC. The pre-fuse must be selected according to the conductor cross-sectional area and the selectivity.

  Additionally observe the national legal regulations!
- The door must be connected to the potential equalisation of the building.
- An order-specific electrical drawing is included with each door. Wire the control system according to this electrical drawing.
- If a residual current device (RCD) is used, an AC/DC sensitive short-time delayed type B RCD is mandatory. Only this type detects AC residual current, pulsating DC residual current and also smooth DC residual current which could be caused by the MCC in the event of an error. In most European countries RCDs with a rated residual current of max. 30 mA are demanded. The use of standard RCDs (type A) is not permissible.

  Additionally observe the national legal regulations.
  If you are in any doubt please consult your electrician.
- When connecting the shielded cables (motor, brake and encoder cable) be ensured that there is good contact between the shield and the metal cable gland. Otherwise the door could malfunction!

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- The main supply cable must have a cross-sectional area of at least 1.5 mm^2 . Conductors with cross-sectional areas up to 2.5 mm^2 can be connected.

4.2.2 Cross-section wiring / plug-in terminals

Cross-section plug X1, X2, X3, X26 and protective earth terminals:

0.5-2.5 mm^2 without wire end sleeve,

0.5-1.5 mm^2 with wire end sleeve without/with plastic collar to DIN 46228-1/-4

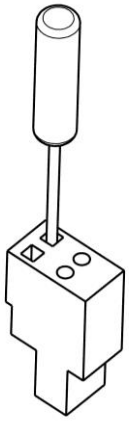
Cross-section plug X4-X21

0.14-1.5 mm^2 without wire end sleeve,

0.14-1.0 mm^2 with wire end sleeve without/with plastic collar to DIN 46228-1/-4

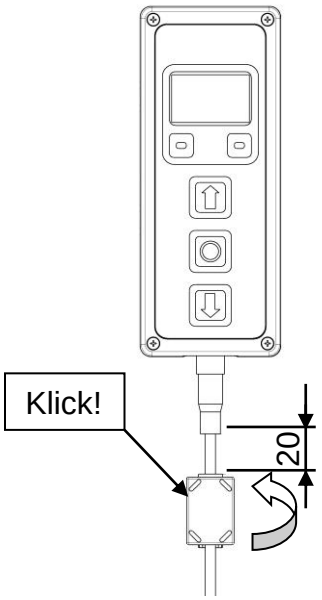
4.2.2.1 Handling recommendation terminal blocks

1. Insert a mini screwdriver (slotted 0.4 x 2.5 mm) in the angular operating slot up to the stop.
2. The screwdriver blade holds the clamping spring open automatically so that the conductor can be introduced into the terminal block.
3. Insert the wire into the terminal block.
4. Withdraw the screwdriver – the conductor is automatically clamped.



4.2.3 Ferrite at the display cable

For EMC reasons it is required for motor cable length greater than 2 m that two ferrites are mounted on the display cable. The Ferrites have to be assembled at both ends at a distance of approx. 20 mm. The assembly at the display side is shown exemplarily. The ferrite are normally pre-assembled or part of the delivery.



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4.3 Wiring restrictions

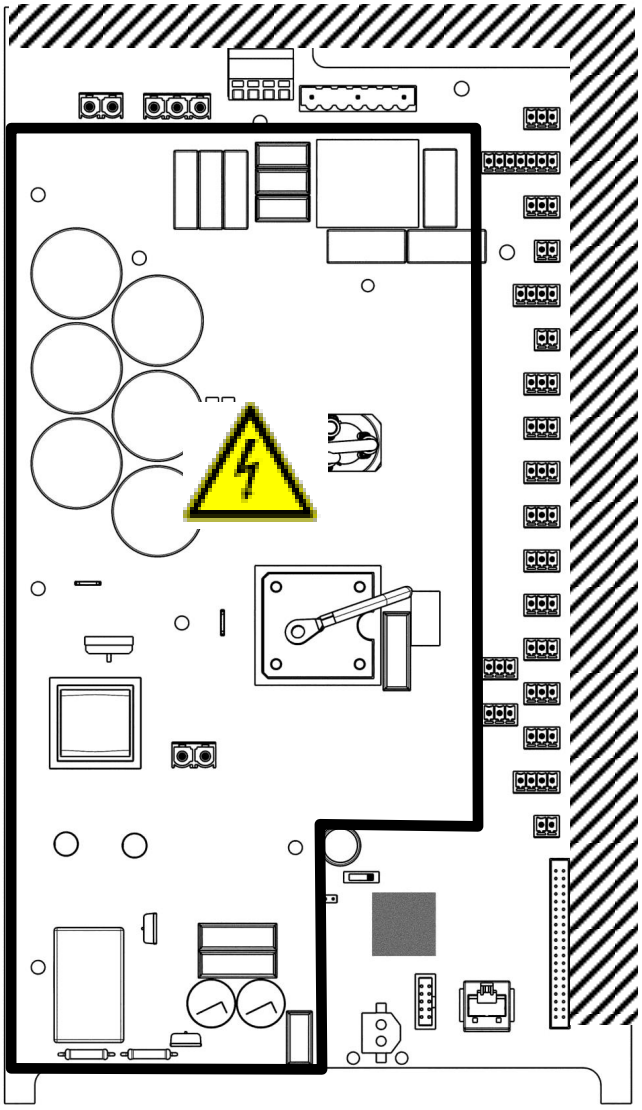
The wiring must be installed in the hatched area to fulfil the requirements for the protection class. For wiring which is designed for an insulation test voltage of 3kV, this restriction is not valid.

DANGER

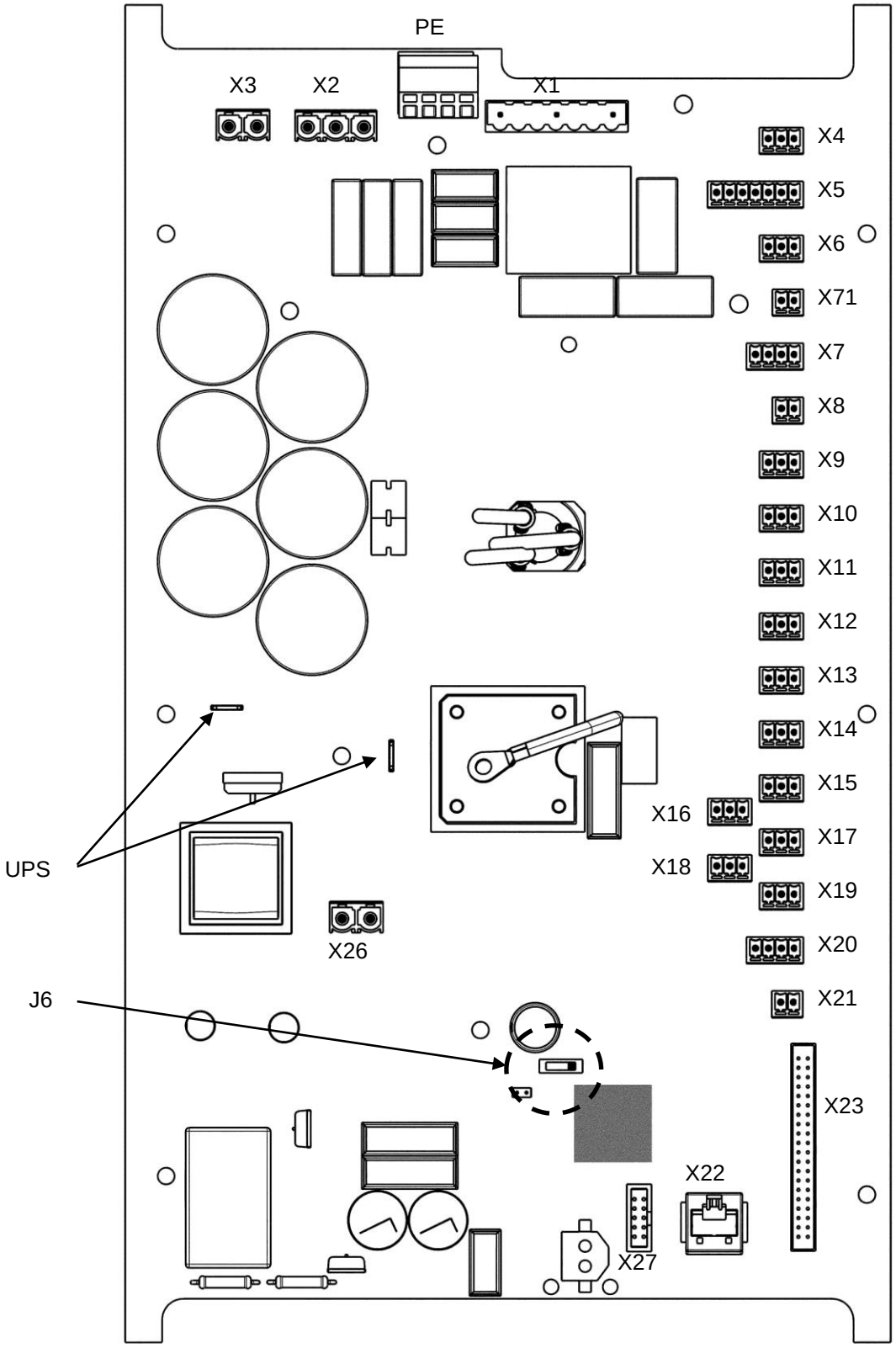
Risk of electrical shock

Dangerous voltage can apply to low voltage wires / signal wires, which can lead to an electric shock with serious injuries or death.

- ⇨ Install wires only in hatched area.
- ⇨ Use only wires with an insulation test voltage of 3kV in other areas.
- ⇨ Bundle wires if necessary.



4.4 Terminal assignment of the control system



WARNING

Emergency stop or other external safety-related signals don´t fulfil any safety requirements if not connected to the terminals X 7 and / or X 71.

Dangerous, not detected failures of the safety devices can occur.

⇨ Emergency stop or other external safety-related signals must be connected to the terminals X 7 and / or X 71.

⇨ Observe these points for design and wiring of external safety devices.

Terminal	Function	Pin assignment
PE	Protective earth connector main voltage and motor cable	
X 1	Main power	1=L1, 4=L2, 7=L3
X 2	Motor connection	1=U1, 3=V1, 5=W1
X 3	Brake	1=BN, 3=BU
X 4	Thermal contact	1=24 V DC, 2=Thermal contact, 3= 0 V
X 5	Encoder	1=24V DC, 2=channel A, 3=0V, 4=zero pulse (not used), 5=24 V DC, 6= channel B, 7=0V
X 6	Reference limit switch	1=24 V DC, 2=reference limit switch, 3= 0 V
X 7	Emergency stop	1/2=emergency stop 1 3/4=emergency stop 2, Emergency hand crank
X 71	Emergency stop	1/2=emergency stop 3
X 8	Emergency manual lever	
X 9	Receiver, door line photocell	1= 24 V DC, 2=photocell, 3= 0 V
X 10	Transmitter, door line photocell	1= 24 V DC, 2=not used, 3= 0 V
X 11	Safety edge 8,2 kΩ	2/3=8,2 kΩ
	Receiver, pre-running photocell	1= 24 V DC, 2=photocell, 3= 0 V
X 12	Transmitter pre-running photocell	1= 24 V DC, 2=photocell, 3= 0 V
X12-X19	General I/O	1=24 V DC or Output 2=Input, 3=0V
X 20	3-pushbutton station (doors without display)	1=24 V DC, 2=open, 3=stop, 4=close
X 21	Status light	1=0 V, 2=24 V DC
X 22	Modbus interface for display and other operating controls and auxiliary modules	
X 23	M8-junction box	
X 26	Break resistor	1/3=180 Ω
X 27	Clone function	
UPS	Connection UPS unit	
J 6	Switch for emergency lever/emergency hand crank contact selection (chapter 4.5)	

4.5 Emergency manual lever / emergency hand crank contact selection

By changing over the switch (while the MCC is switched off) on the main board you can select whether the emergency manual lever or the emergency hand crank (depending on the door type, details in electrical drawing) is controlled by a normally open or normally closed contact.

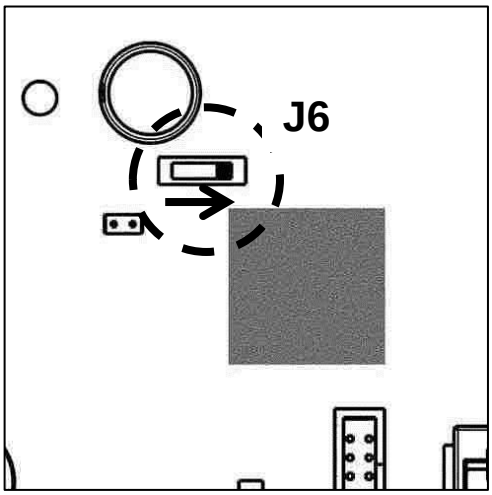
DANGER

Risk of electrical shock

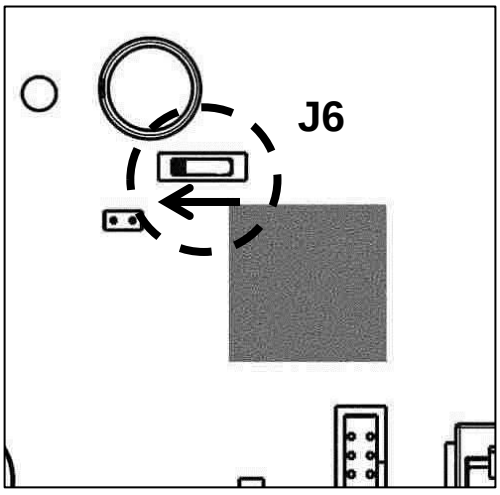
Dangerous touch voltage applies at the control system up to 30 seconds after turning off/disconnecting the supply voltage.

- ⇨ Wait 30 seconds for discharging the capacitors before working at electrical parts and wiring.
- ⇨ Do not touch other than the described components on the board.

The selection will normally be done at the ASSA ABLOY Entrance Systems production. Adjustments are only necessary for replacement units.



NC contact



NO contact

4.6 Reference limit switch

All Albany Doors are supplied with a reference limit switch mounted in the upper end position. This reference limit switch indicates the zero (reference) point for door positioning via the encoder. There are 2 types of reference limit switches:

- Mechanical spindle or cam reference limit switch. This type of limit switch must be set to the upper end position mechanically during installation (see also Assembly Instructions for the door).
- Inductive reference limit switches. This type of limit switch does not need to be set during installation.

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4.7 Setting door limits

Set the end positions before using the door. There are two types to teach-in.

WARNING

Risk of material damage

If the door is moved beyond the lower end position manually during installation (in hold to run mode), this may result in severe damage to the door.

⇨ Observe the door during setting the end positions permanently.

While setting the door limits the door will only move in creep mode in hold to run mode.

4.7.1 Setting door limits for doors without displays

The status light (optional in doors without a display) flashes very fast throughout the entire setup process. This means that no error messages or service information is shown during installation.

To set the final door positions the open and stop buttons on the pushbutton station must be pressed together for approximately 10 seconds. After this the status light will start flashing. The flash codes show which setting must be carried out next.

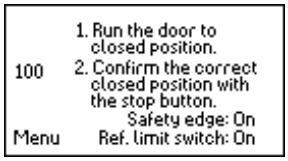
Flash code	Meaning
2 fast flashes	Set the lower end position
3 fast flashes	Set the upper end position (on the reference limit switch)
4 fast flashes	Set the hold-open time

When the appropriate end position has been reached this is confirmed using the stop button on the pushbutton station. After saving the upper end position the hold-open time (4 flashes) can be set. For this wait for as long as the door should remain open in future and then press the close button to save this.

Example:
Hold-open time = 6 seconds => after saving the upper end position (the status light will flash fast 4 times) wait for 6 seconds and then press the “close” button. If the close button is pressed within 2 seconds the hold-open time is switched off.

4.7.2 Setting door limits for doors with displays

The control is delivered without the limit positions set; therefore this view will be the first that appears when the control system is powered up. It is also possible to set the door limit positions by pressing the open and stop buttons on the keypad together for approximately 10 seconds. The following message will then appear:

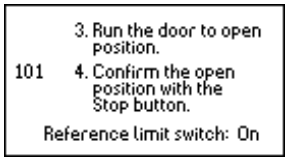


In this view the setup cannot be restarted again through simultaneous pressing of the Open and Stop button. For renewed activation of the setup, press the Stop pushbutton and then restart the setup by simultaneously pressing the Open and Stop button or by switching the Main switch off and back on again.

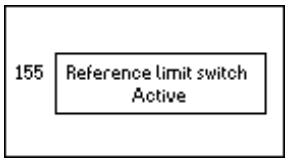
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The Menu button can be used to quit the setup at this point to check or change the default settings (e.g. brake voltage).

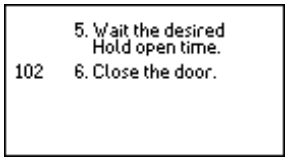
Now use the keypad to move the door to the lower final position and save this by pressing the stop button on the keypad (If the door is already in the closed position, the position can be saved without moving the door.). The following message will then appear automatically.



For doors with a reference limit switch, use the keypad to move the door to the upper end position until it stops automatically. The following message will then appear:



Save the upper end position by pressing the stop button on the keypad. The following message will then appear automatically.

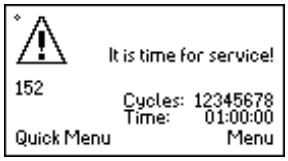


(Be ready for this step it starts once the top position is set!)

The hold-open time can now be set. For this wait for as long as the door should remain open in future. To save this press the “close” button on the keypad.

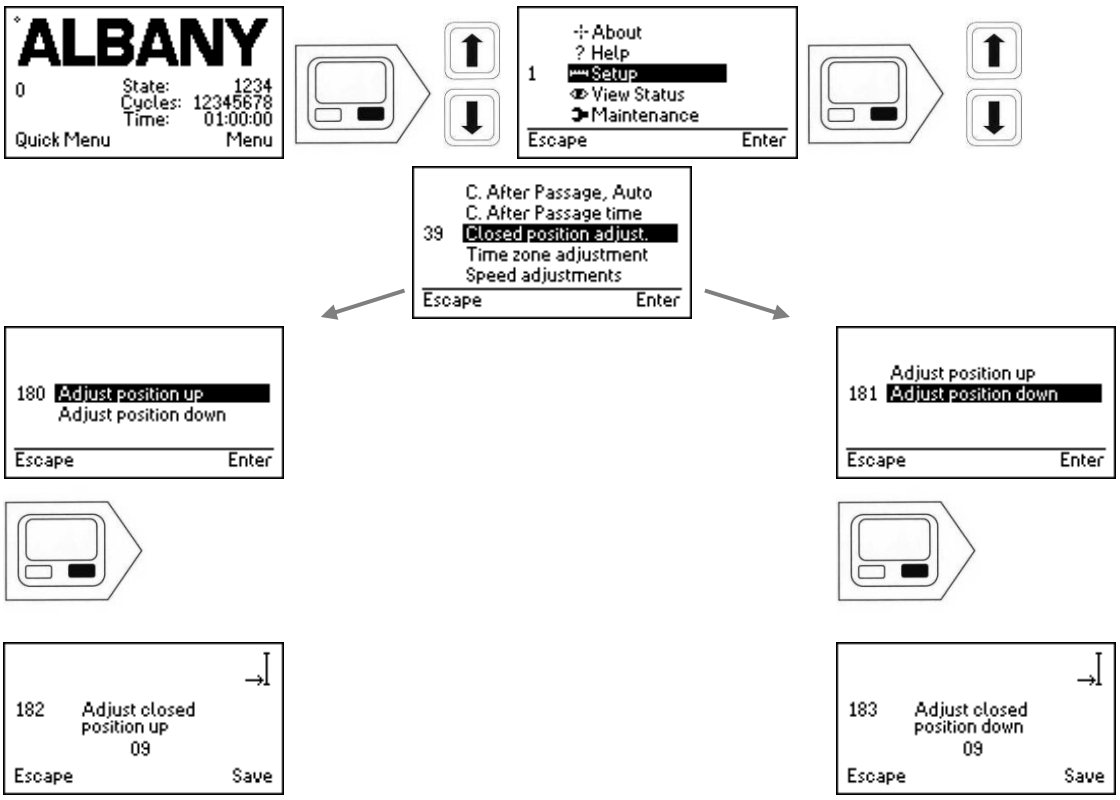
Example:
Hold-open time = 6 seconds => after saving the upper end position wait for 6 seconds and then press the “close” button. If the close button is pressed within 2 seconds the hold-open time is switched off. Instead of the Close button, it is also possible to deactivate automatic closing (hold open time =0) by pressing the Open button. Changing the hold-open time (chapter 4.10)

After the 1st setup the service counter must be reset (chapter 4.33), otherwise the following message appears after several cycles.



This resetting can be omitted in subsequent end position programming.

4.8 Correcting the lower end position [1]

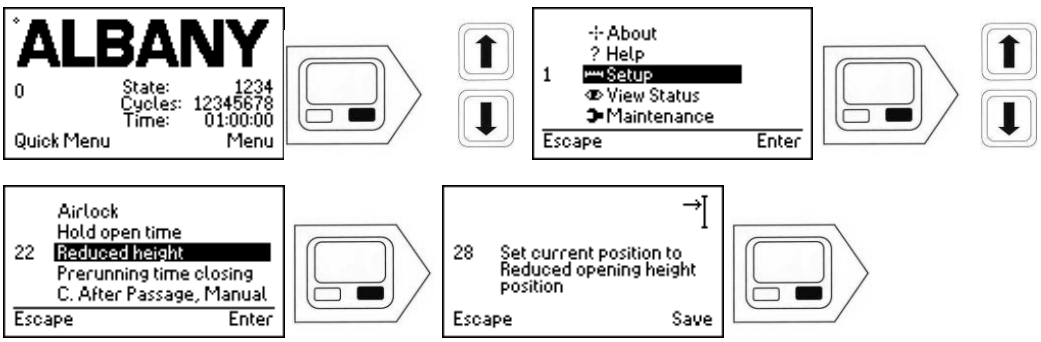


To make sure that the lower end position is not passed the limit should only be changed in small steps! 99 pulses are approximately 1-2cm (0,39”-0,78”). This depends on the door type.

4.9 Setting the reduced opening height [1]

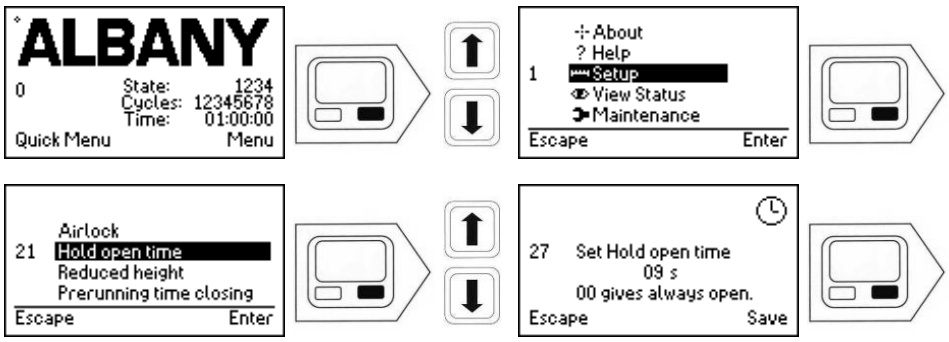
The respective MCC input, which should open the door to the reduced opening height, must be programmed to the function “Reduced opening height” (chapter 4.14).

The door should be moved to the reduced opening height desired. This is then saved as follows:



The reduced opening height is only used for inputs with the appropriately programmed function!

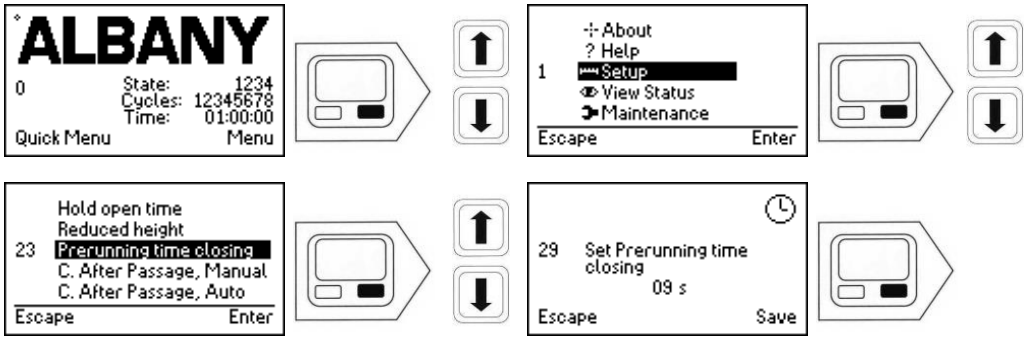
4.10 Setting the hold-open time [1]



This setting is also part of the quick menu.

4.11 Setting the pre-running time before closing [1]

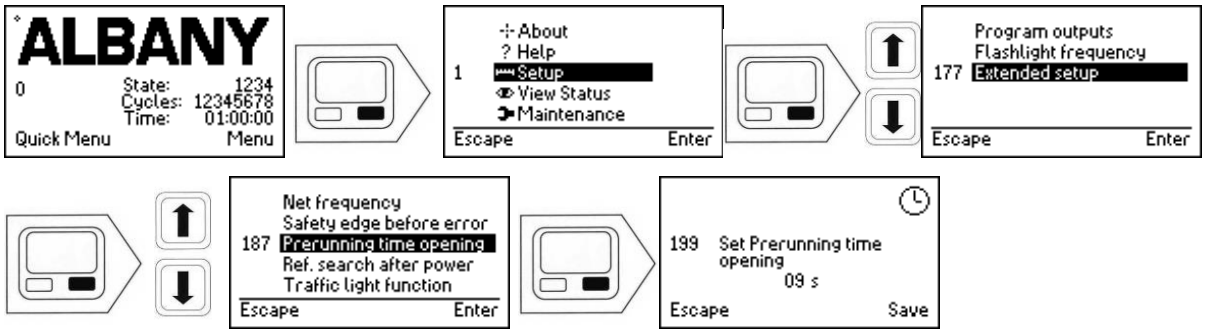
This function is used for traffic light, horn etc. as a warning signal before the door is closing.



This setting is also part of the quick menu.

4.12 Setting the pre-running time before opening [1]

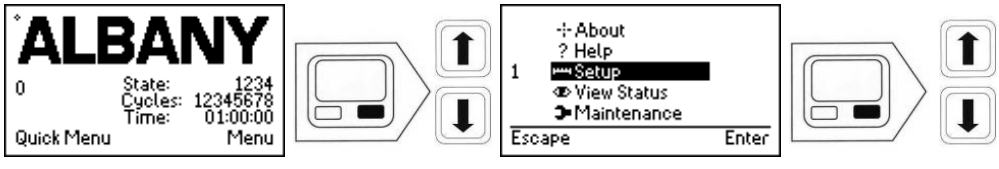
When this pre-running time is activated there will be a delay before the door opens (it opens after the pre-running time).

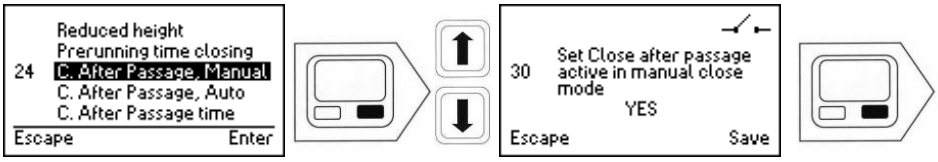


4.13 Close after passage function

4.13.1 Setting the close after passage function without hold-open time [1]

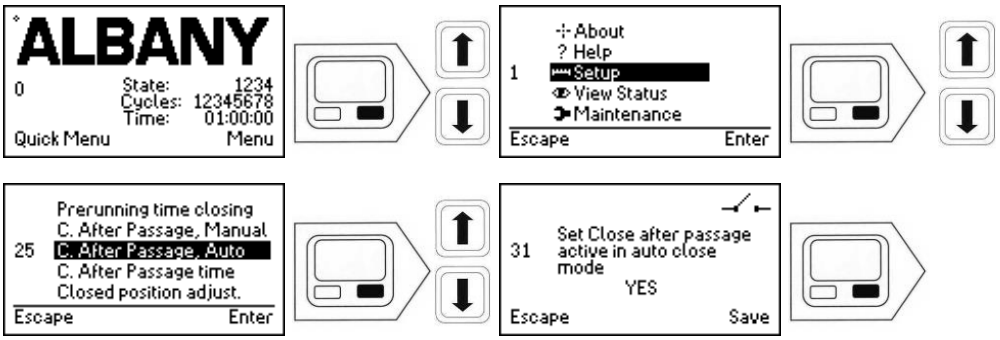
This setting is used to set the close after passage function at actuator inputs with the function “Open command without automatic close function”.



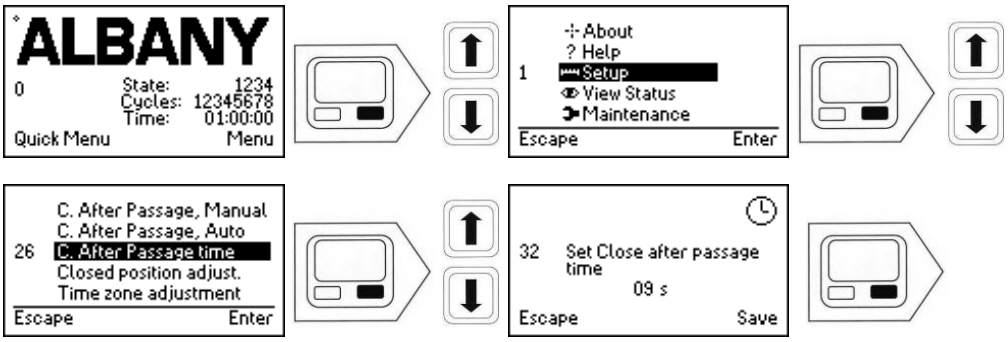


4.13.2 Setting the close after passage function with hold-open time [1]

This setting is used to set the close after passage function at actuator inputs with the function “Open command with closing after time”.



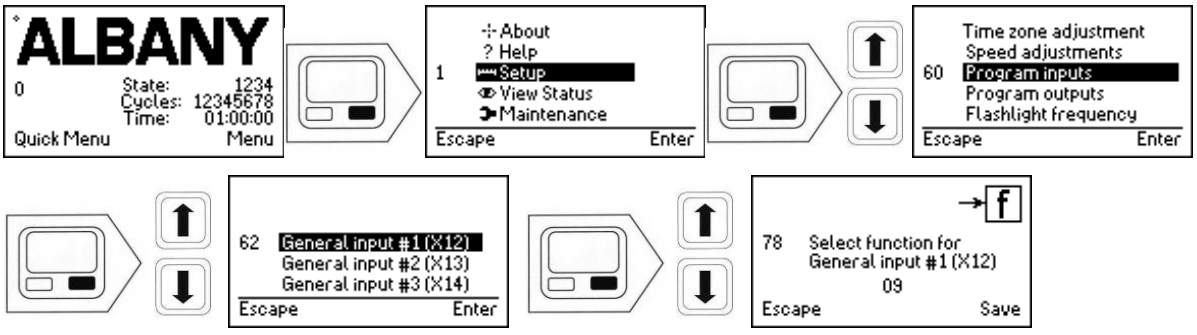
4.13.3 Setting the close after passage time [1]



4.14 Programming the inputs [1]

Select the input to be programmed (observe the electrical drawing) and enter one of the function codes listed below.

For safety reasons inputs 1 and 2 (X12 and X13) can only be re-programmed with system code [3] by the ASSA ABLOY Entrance Systems Service Team.



For traffic lights with recognition of direction it is important that the impulse activators are assigned to the direction. The front is always the side of the door where the display/ control panel is mounted. If direction dependant traffic lights are not used only “front” opening functions shall be used on both sides of the door.

The options are as follows:

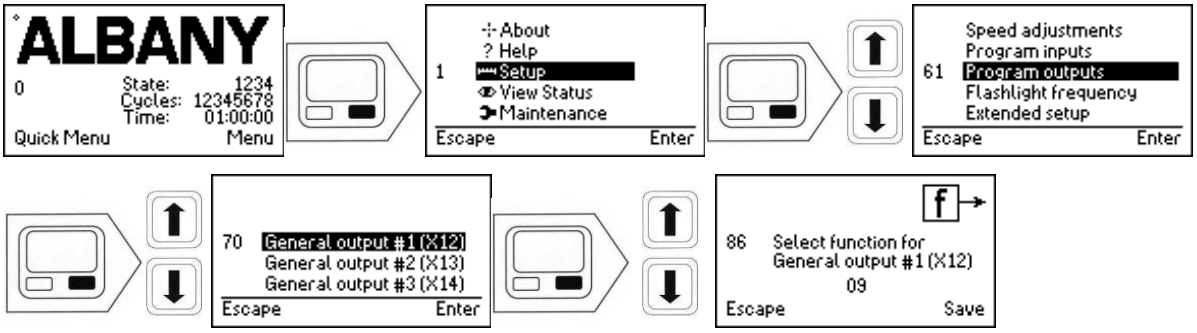
Code No	Function
41	Open command with closure after a time (front)
42	Open command with closure after a time (back)
43	Open command (reduced opening height) without automatic closing function
44	Open command (reduced opening height) with automatic closing function (front)
75	Open command (reduced opening height) with automatic closing function (back)
22	Input without function (e.g. deactivating actuators)
23	Open command (normally open) without automatic closing function
24	Close command (normally open)
25	Stop command (normally closed) => Use Input X7/X71 for emergency stop!
26	Toggle circuit
69	Additional safety (normally closed)
51	Additional safety (normally open)
53	Interlock input (normally closed)
80	Interlock input (normally open)
60	Interlock (switch on/off)
56	Lock door, e.g. mech. door locking, pass door, etc. (normally closed)
54	Crash (normally closed) (function only possible on X12 and X13!)
57	Long vehicle input for airlock (normally open)
45	Middle actuator input for airlock (normally open)
58	Permanent opening display
59	External permanent opening
91	PLC Input 1
92	PLC Input 2
93	PLC Input 3
94	PLC Input 4
95	PLC Input 5
96	PLC Input 6
97	PLC Input 7
98	PLC Input 8
62	Open command (normally closed contact) without automatic close function
71	Breakaway (NC contact) (only for US-market)
67	Input battery signal safety edge – radio system

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4.15 Programming the outputs [1]

Select the output to be programmed (electrical drawing) and enter one of the function codes listed below.

For safety reasons outputs 1 and 2 (X12 and X13) can only be re-programmed with system code [3] by the ASSA ABLOY Entrance Systems Service Team.



The options are as follows:

Code No	Function
65	+24 V DC (continuous voltage)
28	Traffic light (red/green), front (activation relay)
29	Traffic light (red/green), back (activation relay)
30	Switch traffic light on/off with door closed
76	Traffic light green 24 V DC front
77	Traffic light green 24 V DC back
78	Traffic light red 24 V DC front
79	Traffic light red 24 V DC back
63	Output without function (e.g. deactivating activators)
31	Flashlights
33	Interlock
34	Door open
35	Door closed
36	Door ready for work
81	Malfunction indication
37	Status light
38	Pre-running photocell transmitter
40	Additional security
83	PLC output 1
84	PLC output 2
85	PLC output 3
86	PLC output 4
87	PLC output 5
88	PLC output 6
89	PLC output 7

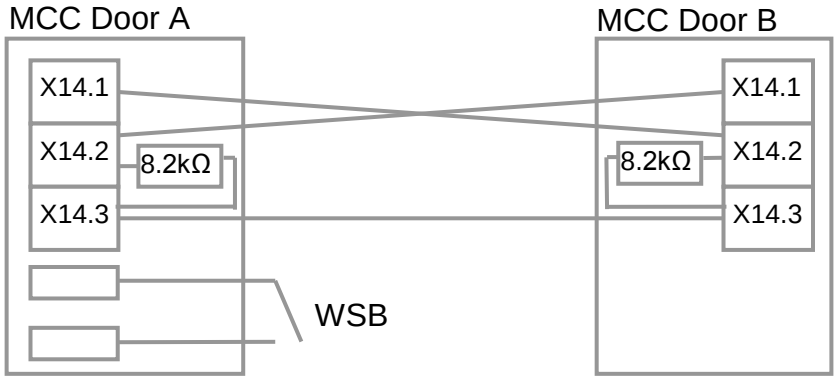
90	PLC output 8
72	UPS On/Off control system

4.16 Airlock function [1]

4.16.1 Airlock wiring

A 3-wire connecting cable is required for the airlock wiring between Door A and B. Two 8,2kΩ resistors are also necessary. The airlock connection of both MCC´s is only possible at the terminals X14!

The following connections are to be provided (Observe electrical drawing of the door!):

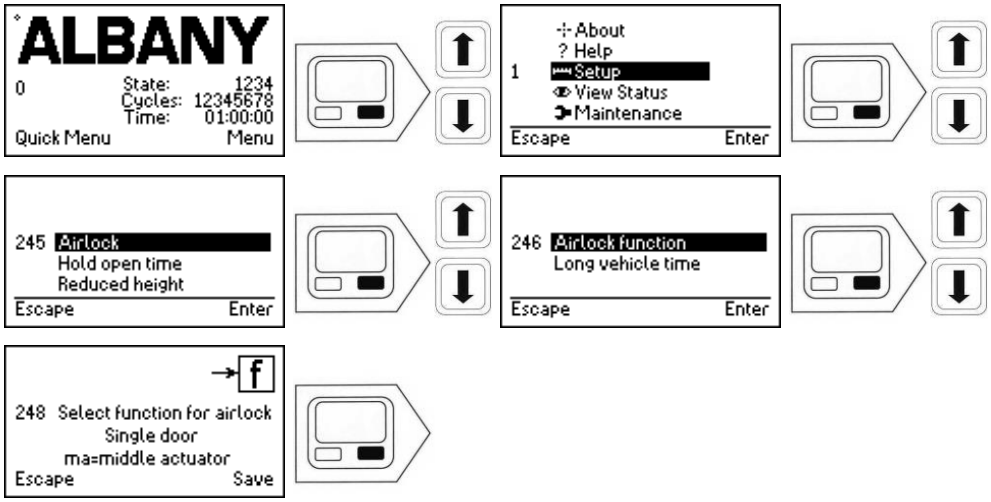


The door with the middle actuator (WSB) is always Door A! The free inputs X12-13, X15-19 can be used.

4.16.2 Setting the airlock function [1]

The functions of the X14 input must be programmed to code number 22 (input without function). The output X14 must be programmed to code number 65 (24 V continuous voltage).

The close after passage time can at wish be activated for actuator with hold open time and for actuators without hold open time (chapter 4.13.1 / 4.13.2).
In this menu item it is possible to select whether a single door, Door A or B of the airlock is concerned and whether the airlock is operated with or without middle actuator.

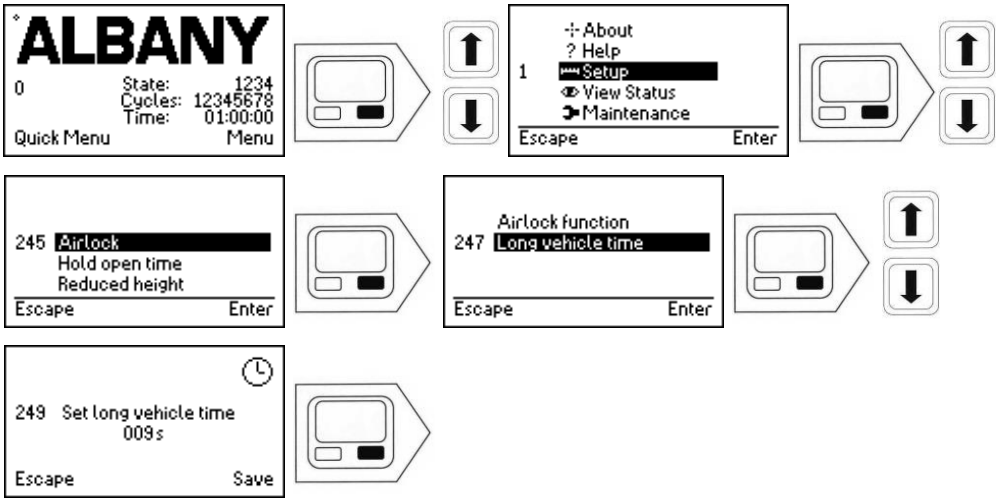


The selection options are:

Function	Description
Single door	No airlock operation
Airlock with middle actuator Door A	Airlock operation with middle actuator Door A
Airlock with middle actuator Door B	Airlock operation with middle actuator Door B
Airlock without middle actuator Door A	Airlock operation without middle actuator door (A)
Airlock without middle actuator Door B	Airlock operation without middle actuator door (B)

4.16.3 Setting the long vehicle time [1]

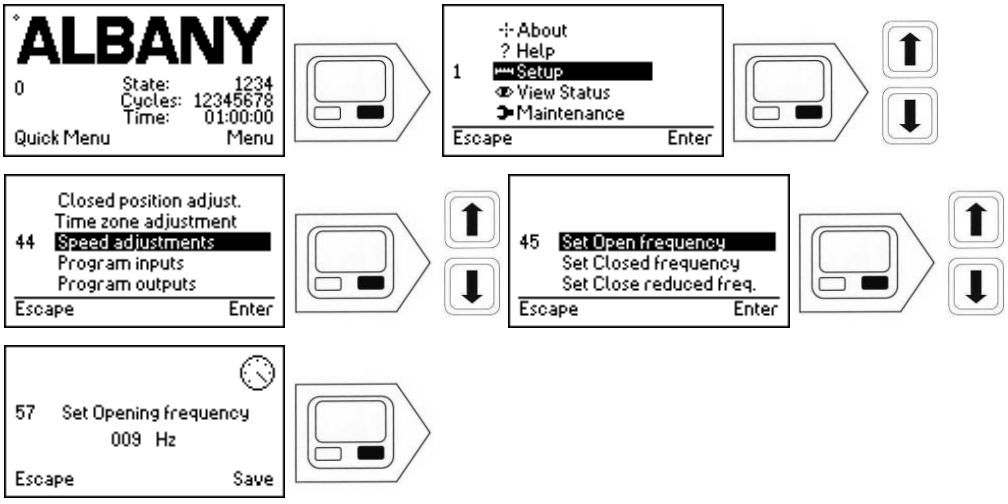
The long vehicle time is the time which the door line photocell in the side frame and the middle actuator have to be simultaneously activated to start the long vehicle function. With the long vehicle function both doors open irrespective of the airlock sequence. The factory setting is 15 s.



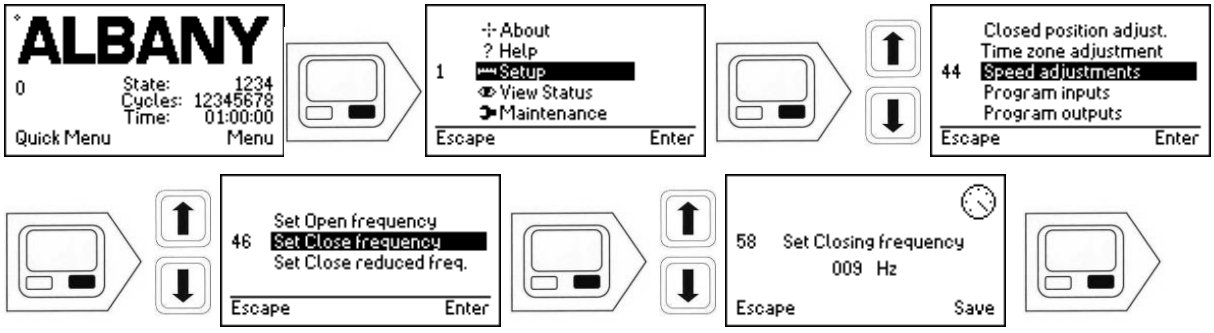
4.17 Reducing the speed [2]

For safety reasons the speeds can only be altered within a fixed range.

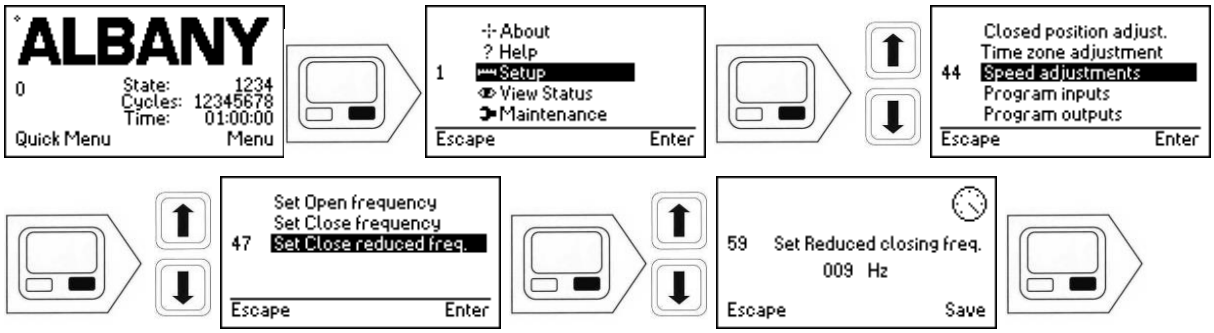
4.17.1 Reducing the opening speed [2]



4.17.2 Reducing the closing speed above an opening height of 2.5 m [2]



4.17.3 Reducing the closing speed below an opening height of 2.5 m [2]



4.18 Adjustment of activations of the safety edge or pre running photocell [2]

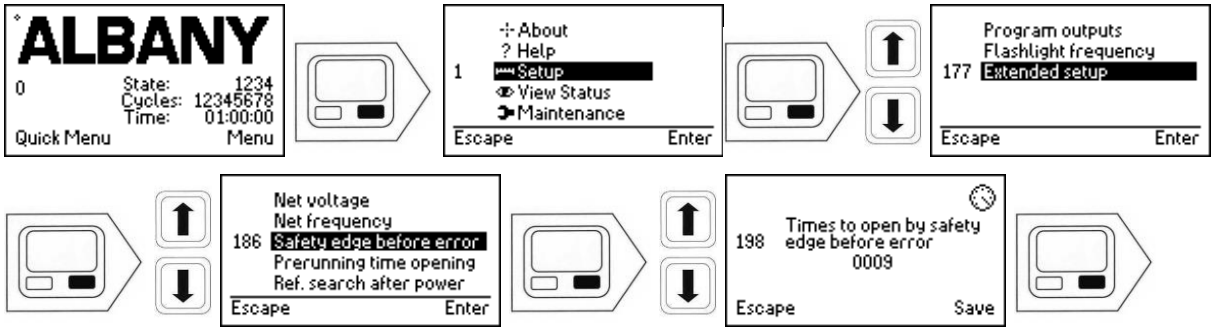
The electrical safety edge serves as closing edge protection. As standard, the door is usually locked in the upper end position after the electrical safety edge has activated once. Using this menu the value can be increased, for example to twice. This means that the door would only lock in the upper end position after two activations of the electrical safety edge (without the door closing completely). For the pre-running photocell this value is set to 10 activations as standard.

WARNING

Personal risk at the door is increased

A collision with the door curtain can be repeatedly performed with increasing the value

- ⇨ The value should only be altered by our service team in reasoned, exceptional cases.
- ⇨ For safety reasons the value for the electrical safety edge should only be increased to a maximum of 3 activations!



4.19 Activating reference search after a power failure [2]

Here it can be select whether a reference search must be performed after a power failure. This monitoring is intended to last for around 5-6 hours without power. Afterwards the reference search must be performed.

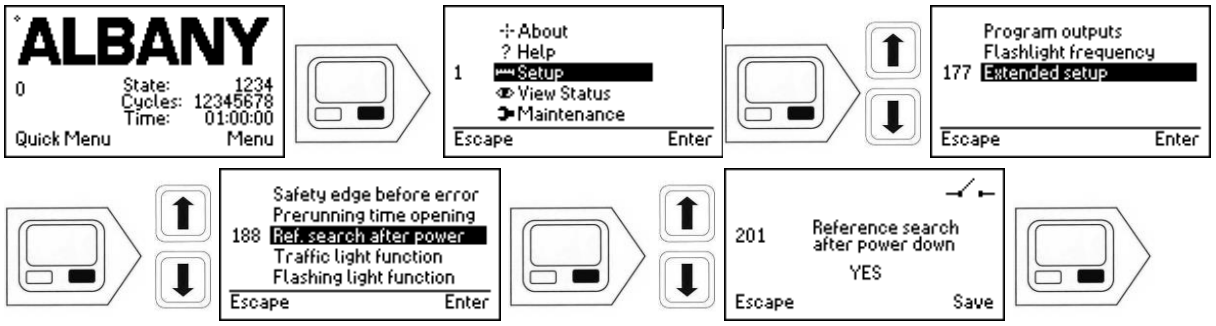
WARNING

Risk of material damage

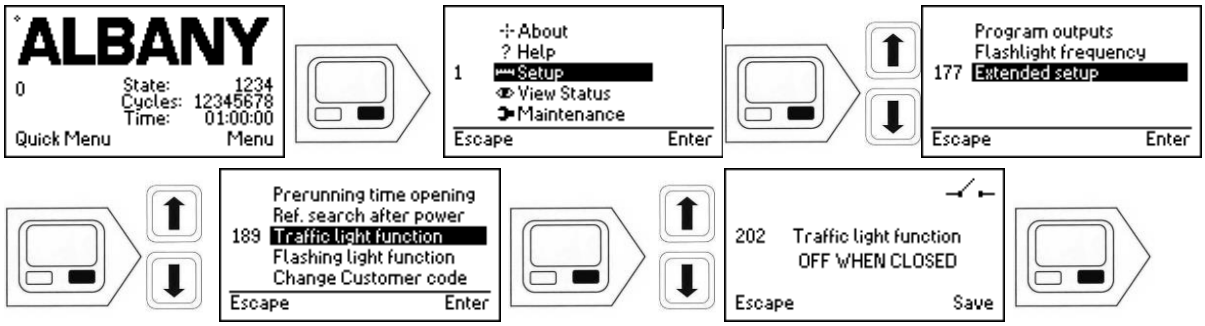
At doors without brake release monitoring (brake release with mirco-switch) the release of the brake while the MCC is switched off (e.g. after a power falure, etc.) can cause an over-running of the door end positions.

⇨ Activate this function only for doors with monitored brake release.

If the emergency manual lever or emergency hand crank monitoring is activated during a power failure, reference search must be performed.

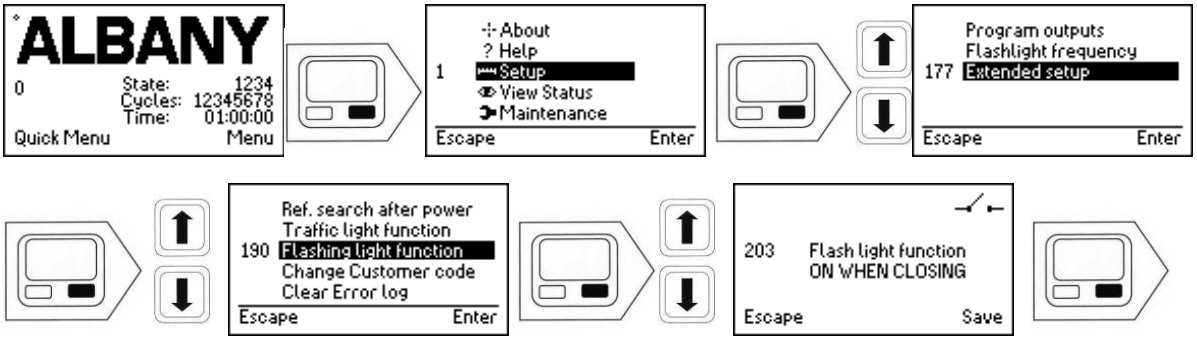


4.20 Altering the traffic light function [1]



Here it can be selected whether the traffic light remains on all the time or switched off when the door is closed.

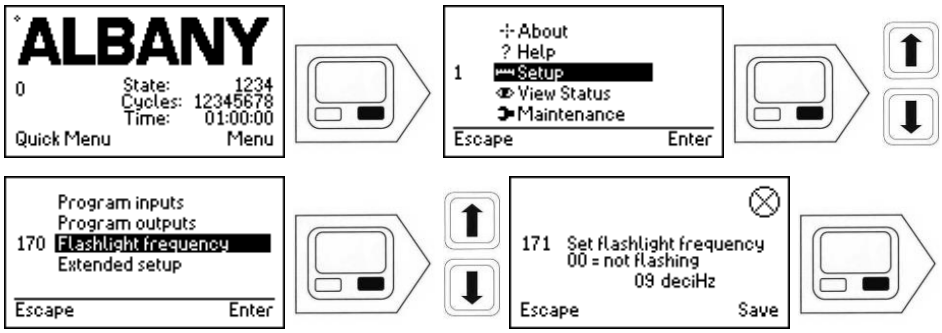
4.21 Altering the flashing light function [1]



Here it can be selected whether the flashing light is on whenever the door moves or only during closing.

4.22 Setting the flash rate of the flashlight [1]

The control system offers the option of using a normal indicator light as a flashlight. For this the switching frequency of the flashlight output can be changed using the following menu option.

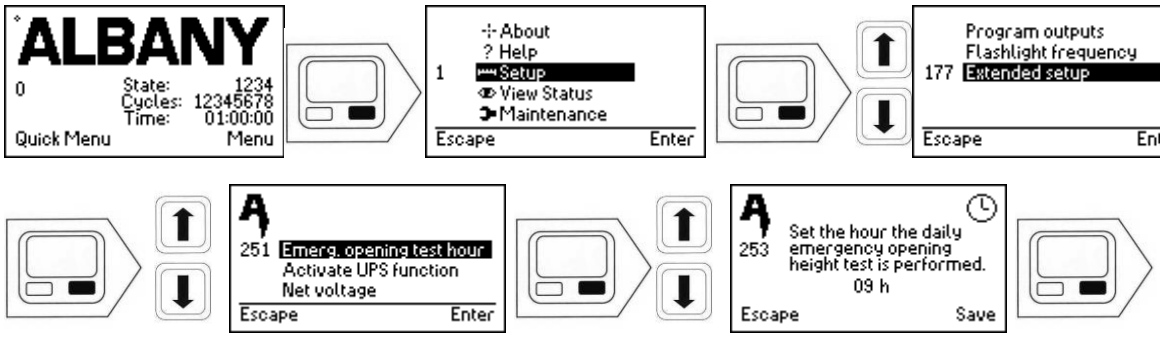


4.23 Emergency exit doors – set time for regular self-test [1]

To ensure the exit and escape route functions, self-tests are performed by the control system after switching on, after the escape route deactivation has ended, after 24 hours and after a power failure.

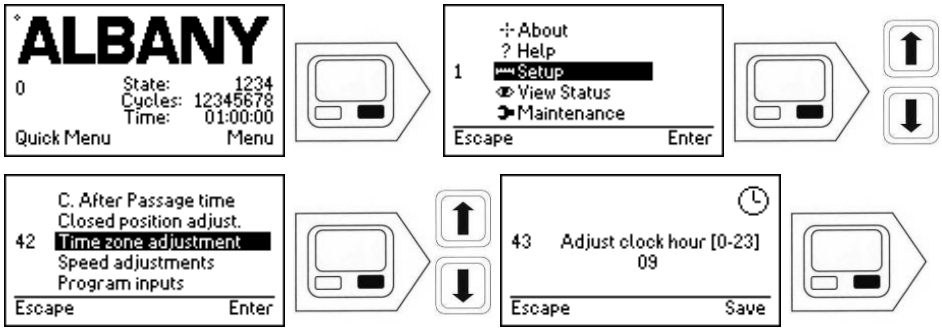
When the self-tests are performed, all safety relevant electrical and mechanical functions of the door are checked.

8:00 am is pre-set in the factory as the time for performance of the self-tests after 24 h. This time can be changed to the user's needs using the display unit.



4.24 Setting the time zone [1]

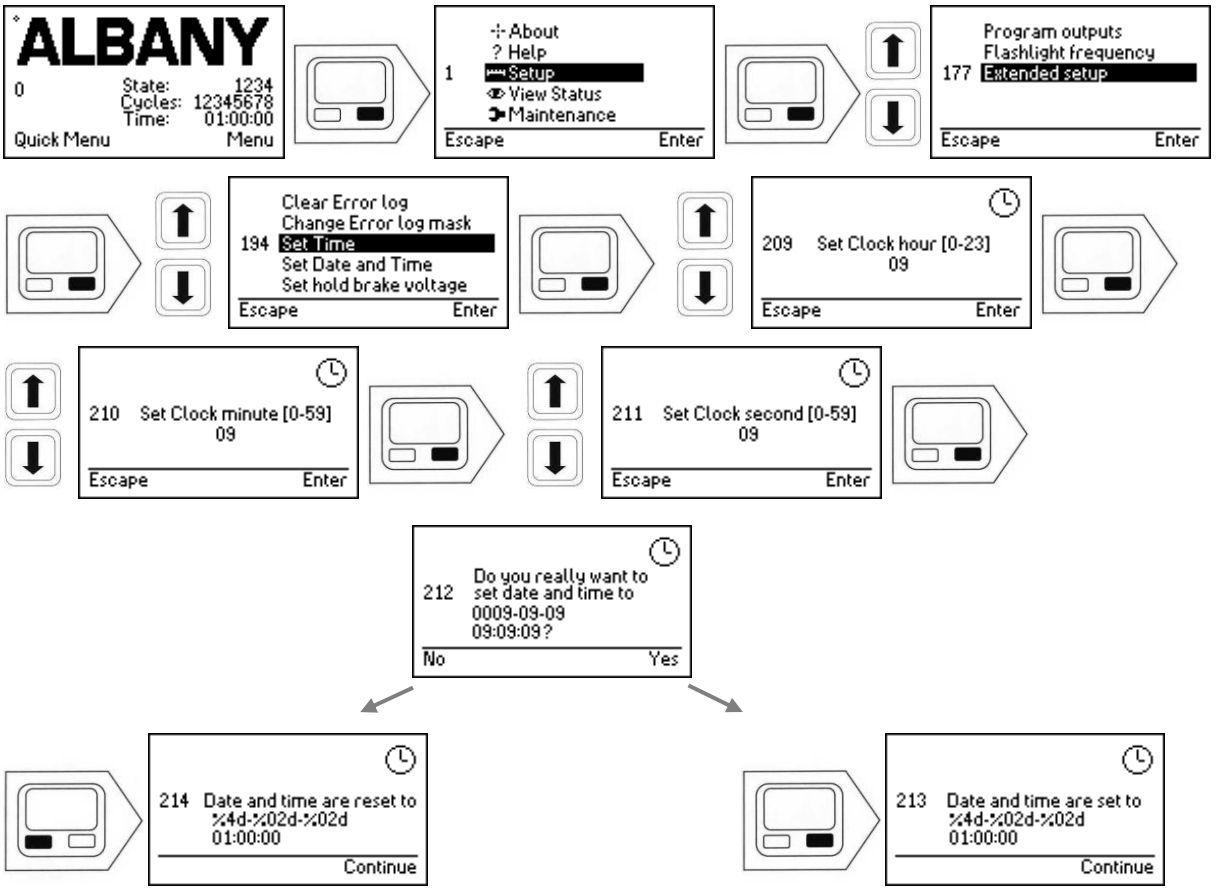
If the door is placed in a different time zone it is necessary to change the clock hour.



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4.25 Setting of the time [1]

If the clock setting is not correct, it is necessary to change the time.



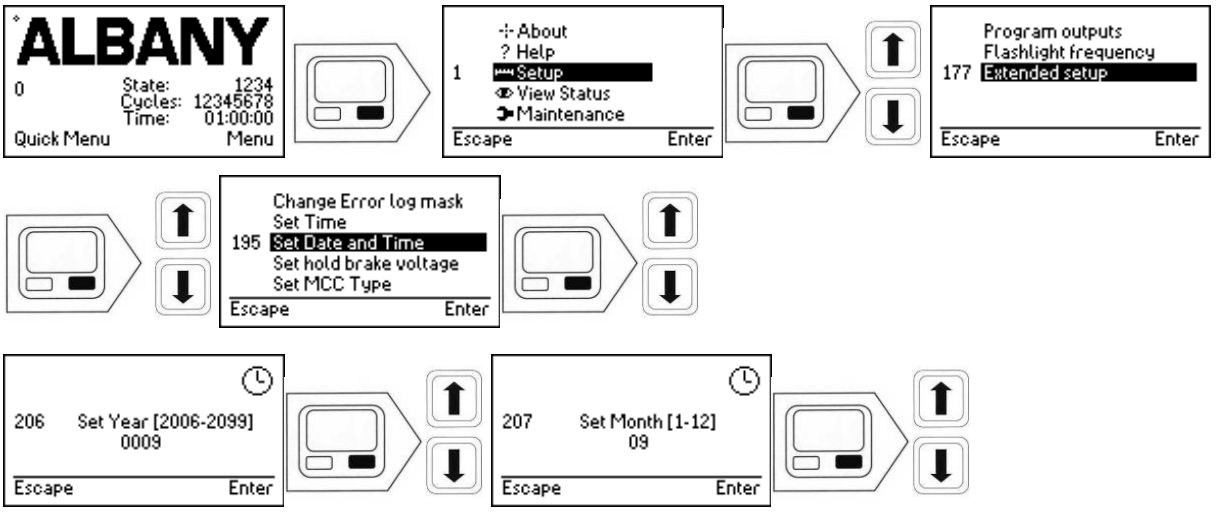
If the adjusted values are not valid following view will be shown.



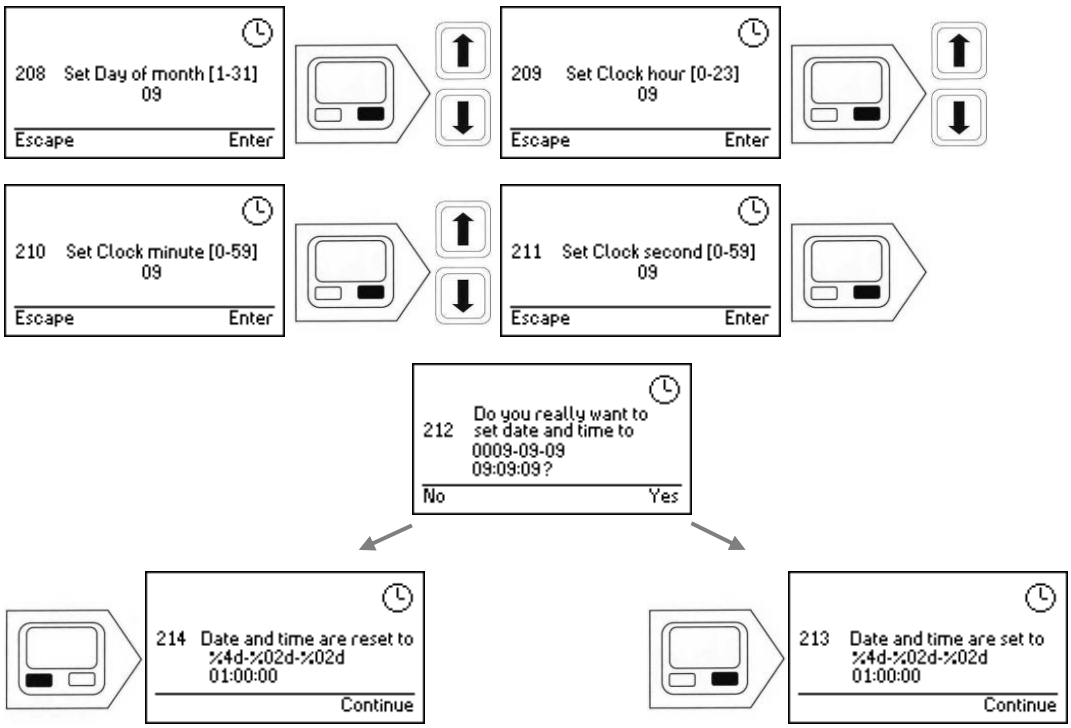
Date and time will be set to the old values.

4.26 Setting of the date and time [2]

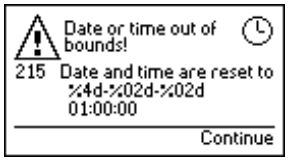
Here the control system date and clock can be corrected.



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If the adjusted values are not valid following view will be shown.



The time will be set to the old values.

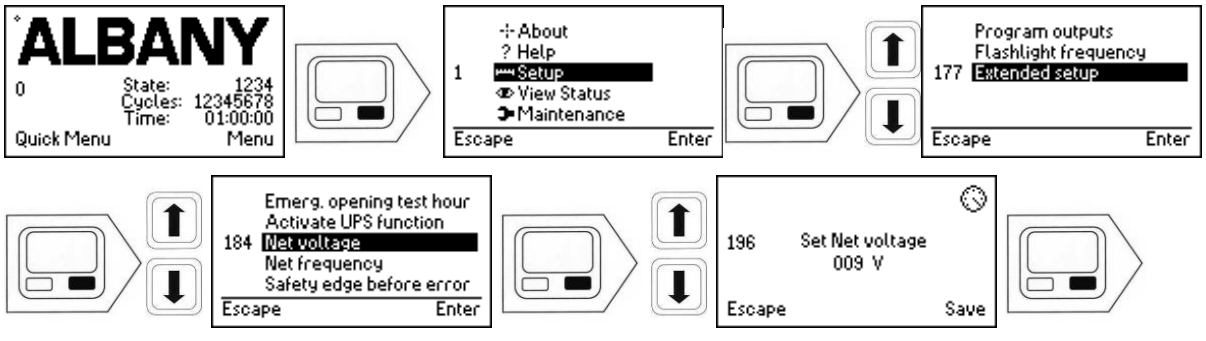
4.27 Changing the main or supply voltage [2]

CAUTION

Risk of material damage

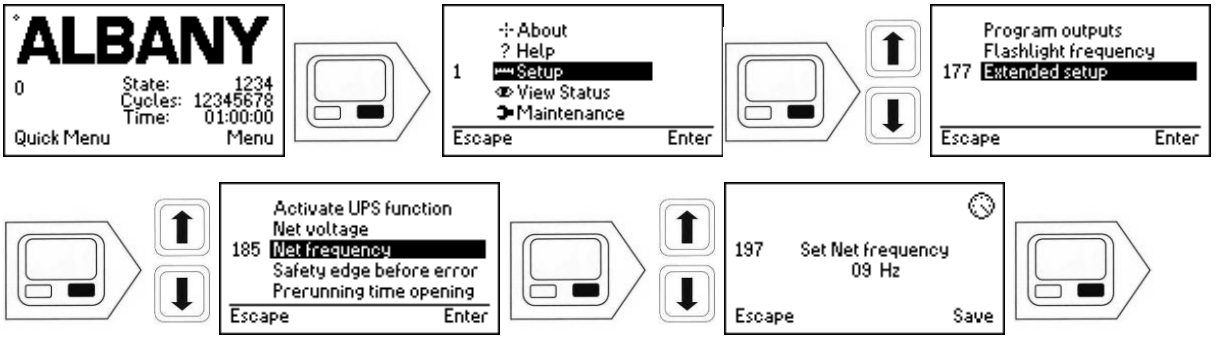
A wrong main voltage setting can destroy the control system.

⇨ Check local main voltage if necessary and adjust accordingly in the MCC.



The control system must be switched off and on again, so that the main voltage change becomes effective.

4.28 Changing the main supply voltage frequency [2]



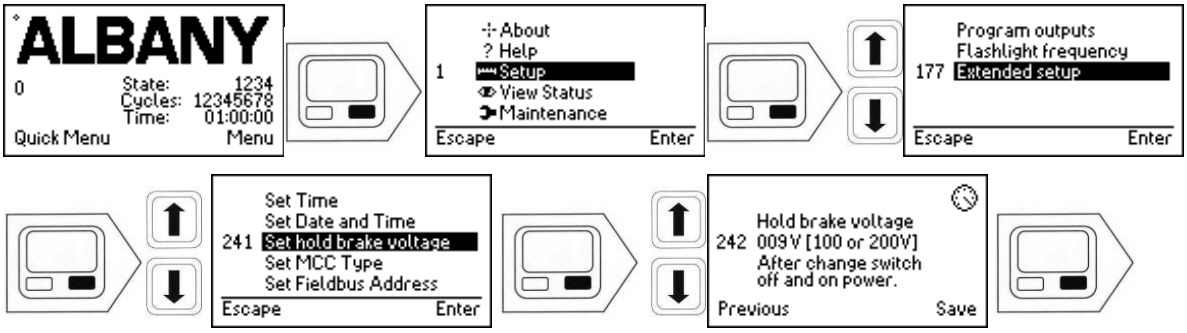
4.29 Changing the brake voltage [2]

CAUTION

Risk of material damage

A wrong brake voltage setting can destroy the control system and the brake.

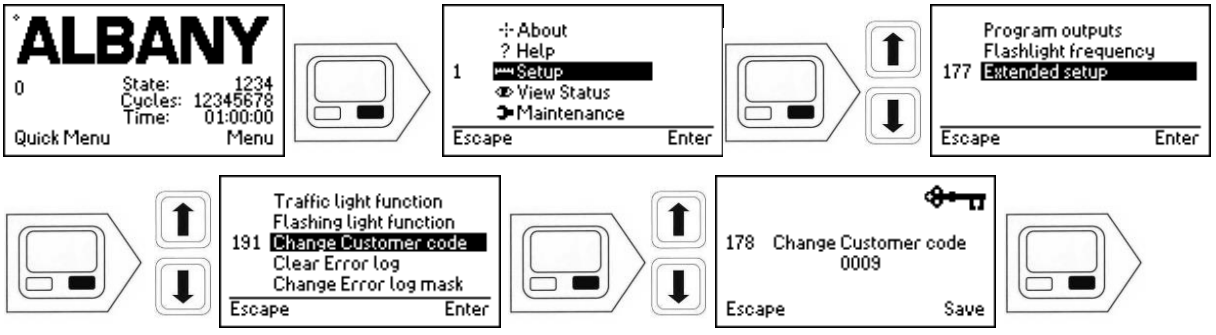
⇨ Observe the information of the type label of the brake.



The control system must be switched off and on again, so that the brake voltage change becomes effective.

4.30 Change customer code [2]

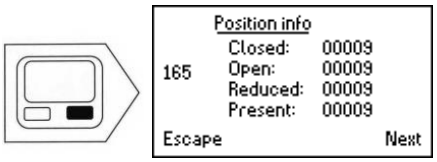
In order to prevent that the user locks their selves out, the code can only be changed by the service technician.



4.31 Querying the door positions and the state history [0]



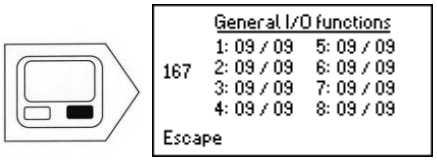
The set date, set time, current MCC temperature, the current software state and the ready for work signal can be read off in this view.



The number of encoder pulses for the end positions closed, opened and reduced opening height and the current position can be read off in this view.



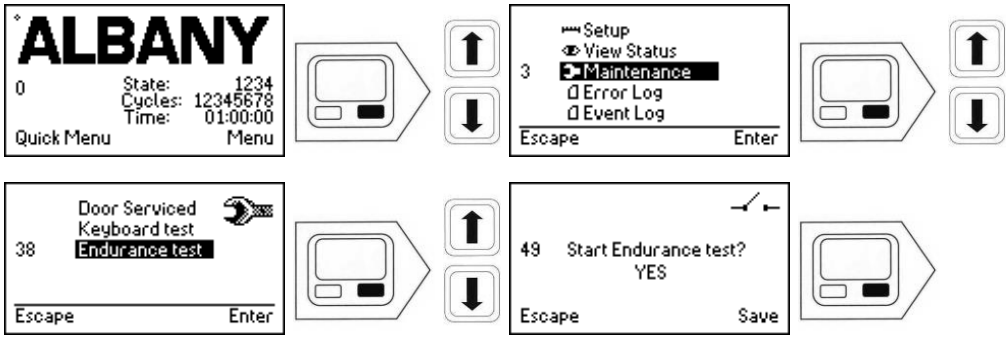
The last 10 and the current state are displayed in this view.



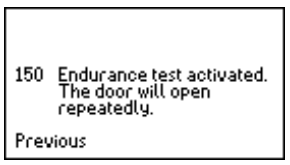
The functions of the programmed inputs and outputs are displayed in this view (for meaning of the codes, see chapter 4.14 / 4.15).

4.32 Setting endurance test [2]

This function should only be activated for test purposes.



Activated continuous operation is indicated by the following message.

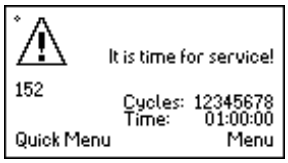


Deactivation is performed in the same order!

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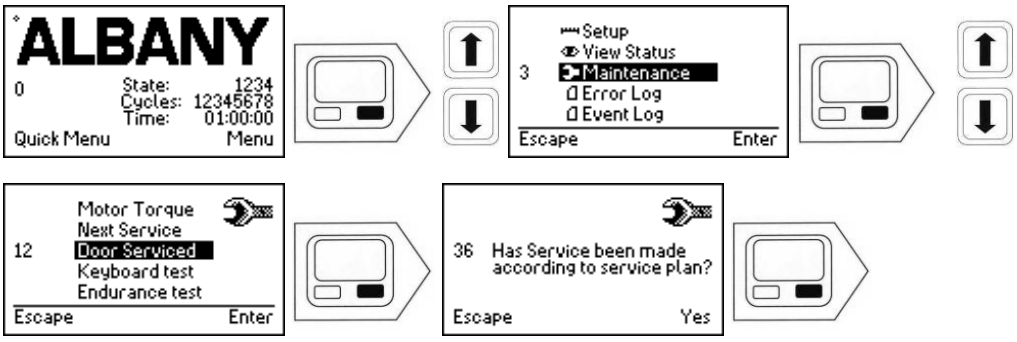
4.33 Reset service interval [1]

If service is due, it is automatically shown with the following message.



Every service interval indicated is recorded in the event log.

If the service has been carried out in accordance with ASSA ABLOY Entrance Systems specifications the service interval counter must be reset. Every time the service interval counter is reset this is recorded in the event log.



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5 Operation

5.1 Safety instructions

Observe the safety instructions in Section 2 and the safety instructions in the user guide for the door.

5.2 Display

The display shows messages, faults, etc. in plain text or graphically. During operation the time, cycle counter and status information are shown.

5.3 Switch-on and reference run

The position of the door and the vector control of the motor are regulated by an incremental encoder. This encoder is firmly mounted on the motor. A reference search is therefore necessary when the system is switched on for the first time. If for doors with brake release monitoring (normally doors with emergency manual lever) the control system is switched off after the reference search, a renewed reference search is only necessary if the emergency manual lever has been used (alteration of the door position).

This monitoring is intended to last for around 5-6 hours without power. Afterwards the reference search must be performed. Refer to the following flow diagram for this. A reference search is always necessary for doors without brake release monitoring (normally for doors with hand crank).

Observe the information and setting options for this function in chapter 4.19.

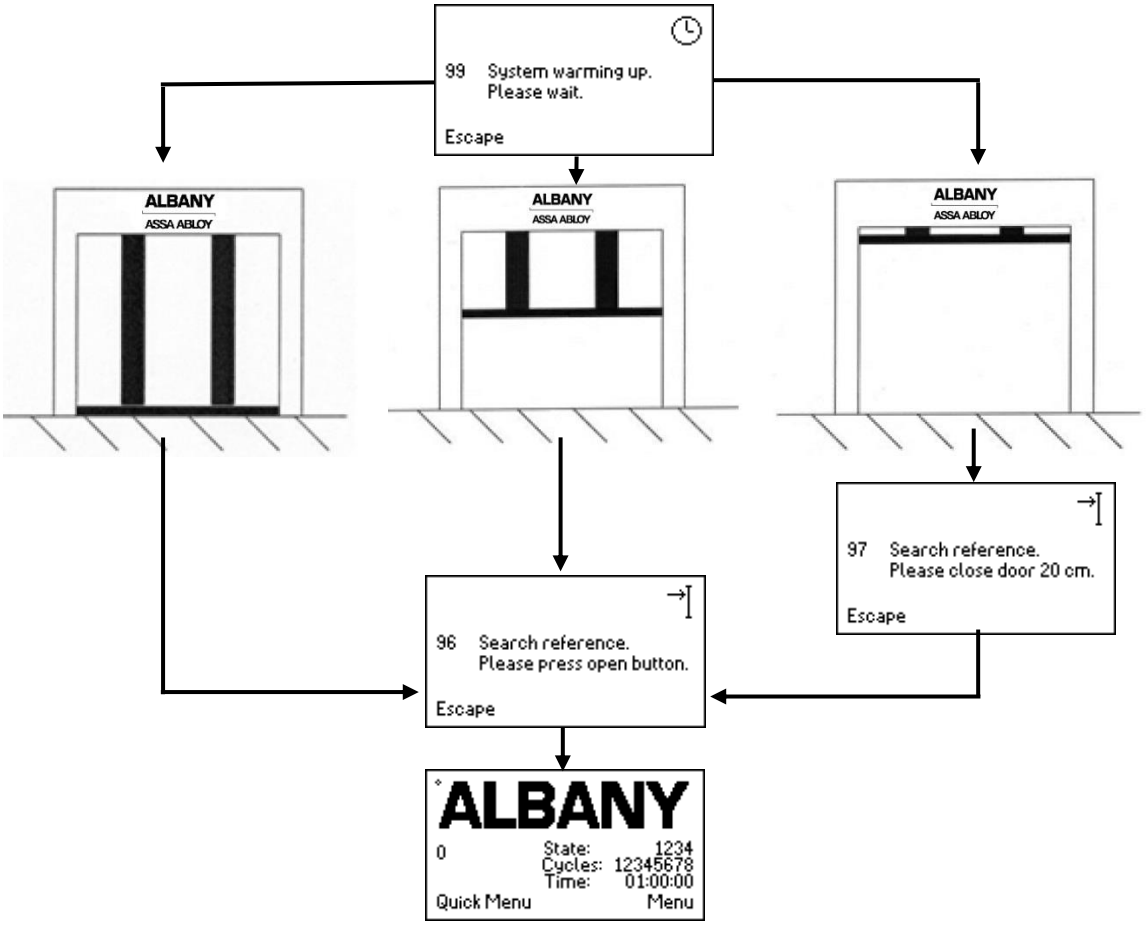


The reference search is initiated automatically by pressing the “Open” pushbutton or from external actuators (e. g. motion detector).

If the main switch is switched on, and if the emergency manual lever has not been activated since the door was last switched off (maximum 5-6 hours) a reference search does not need to be performed. After an open or close command the door will run in standard operating mode.

If the emergency manual lever has been used, or the maximum monitoring time of the emergency manual lever has been exceeded a reference search must be performed.

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The door is now ready for operation.

5.4 Opening the door

An “open” impulse is given by any impulse activator or from the keypad. The brake is released; the drive motor is activated by the MCC with a pre-defined acceleration ramp. The door opens with maximum speed until it reaches the starting point of the upper deceleration ramp. The door is now decelerated by the frequency converter and stopped in the adjusted upper end position. When this position is reached the motor is switched off and the brake is activated.

Depending on the function, which has been pre-selected, either the hold-open or close after passage time is started or the control system waits for a closing impulse.

5.5 Closing the door

Once the end of the hold-open time or close after passage time is reached or a closing impulse is received all connected safety devices are first tested by the control system. If this safety test is successful the safety devices are activated, the brake is released and the motor is driven by the frequency converter with a pre-defined acceleration ramp. The door closes as far as the starting point of the lower deceleration ramp. The door is now decelerated with a pre-defined deceleration ramp and stops in the lower end position. When this position is reached the motor is switched off and the brake is activated.

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The safety devices are switched off max. 50 mm (2”) before the lower end position is reached. This prevents the door from reversing when it reaches the ground.

5.6 Stop the door

The door can be stopped in any position by pressing the “Stop” button. An “Open” or “Close” impulse then starts the door’s operation again. If the stop button on the display unit is pressed it is necessary to press the display open or close button to start the door. The control system will also remember that it is stopped with the display stop button if it is restarted by switching on and off the main supply, also in this case only the display open or close button can start the door again.

5.7 Power failure

If a power failure occurs doors with power off brake immediately stops wherever it is at that moment. Once the main power returns the door will only be in normal operation after a renewed impulse.

Doors with power on brake open or close automatically in the event of power failure depending on the spring balancing and door position. A closed door will open and an open door can close slightly (depending on the spring balancing).

5.8 Emergency manual operation

If there is a power failure or repair work or maintenance is being carried out it may be necessary to open the door manually.

Doors with power on brake do not have an emergency manual lever!

WARNING

The door may partially open or close by itself after the emergency manual lever has been activated!

There is the risk of injuries by the movement of the door curtain.

⇨ Observe the door during use of the emergency manual lever permanently.

WARNING

Risk of injury

It can lead to injuries by simultaneous starting of the motor during inserting the emergency hand crank.

⇨ Always switch off the main switch before using the emergency hand crank.

By using the emergency manual lever on the side frame the brake of the drive motor can be released via a disengagement cable. Depending on the balancing system and the door position the door curtain may partially open or close by itself. Now it is possible to reach below the bottom rubber in the middle of the door and push the door curtain up. While doing this the door curtain will wind onto the top roll. To get back to the normal operation of the door the emergency manual lever should be returned to its original position. The control system can then be switched on and the door can be operated normally.

At doors with emergency hand crank, the door can be opened and closed by turning the emergency hand crank which must be inserted on the bottom of the motor.

If the emergency manual lever is still operated when the main switch is switched on the following message appears.

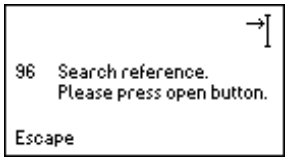
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The following message appears for doors with emergency hand crank



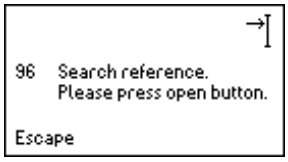
This message is cancelled automatically once the emergency manual lever has been returned to its original position or the emergency hand crank has been removed. If then a reference search is needed, the following message will then appear:



The reference search is initiated by pressing the “Open” pushbutton (chapter 5.3). The door is then once again ready for work.

The door is not designed for regular manual use.

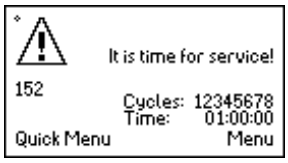
For doors with power on brake, the brake automatically releases if the main voltage is switched off, e.g. by switching off the main switch or disconnecting the CEE plug. A reference search must be initiated after the main voltage has been restored. Therefore the following message appears automatically:



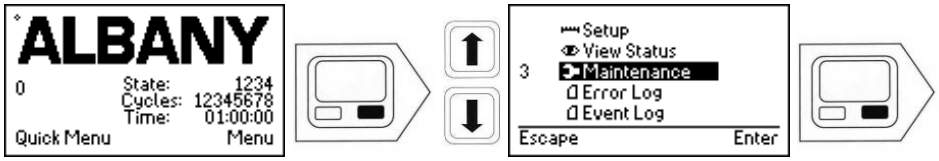
The reference search is initiated by pressing the “Open” pushbutton (chapter 5.3). The door is then once again ready for work.

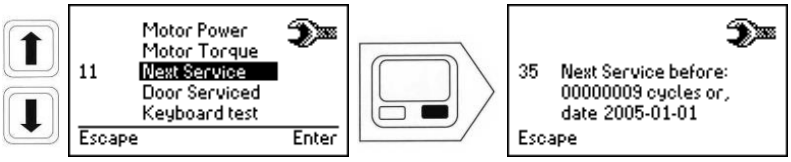
5.9 Service interval

If service is due, it is automatically shown with the following message. Every service interval indicated is recorded in the event log.



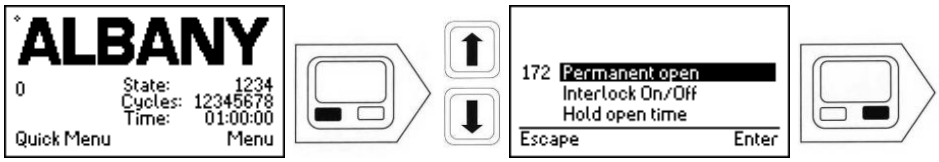
The date when the next service is due can be called up on the display.





5.10 Permanent opening of the door [1]

Permanent opening can be activated as following.

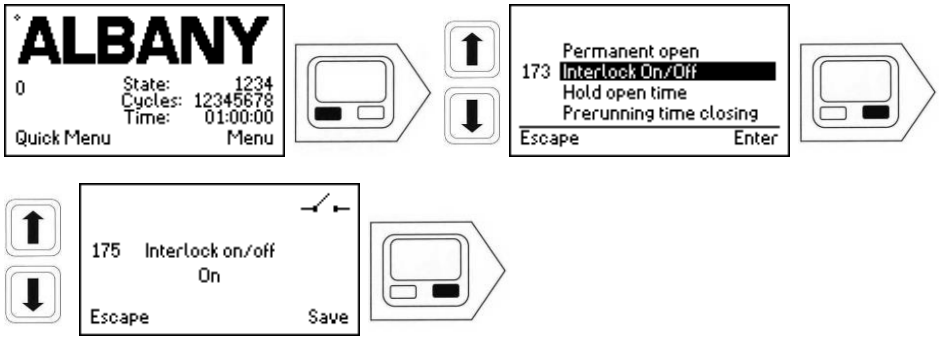


Permanent opening can be deactivated again by following the same sequence.

5.11 Deactivation of Interlock function [1]

Doors that are interlocked with one or more other doors have the option of deactivating the interlock function. This could be necessary if maintenance work is being carried out on one of the doors or the door shall be used as a single non-interlocked door, for example in summer.

The interlock function can be reactivated by following the same sequence.



5.12 Crash-system

In order to minimise the damage of the door in case of impacts doors can be equipped with a crash-system. If there has been an impact to the door and the crash sensor has been activated the door is stopped immediately. The following error message appears.



WARNING

Risk of personal injury and material damage

Damaged parts of the door can endanger people or cause associated consequential damages.

⇨ After any impact the door should be checked thoroughly for possible damage.

5.12.1 Auto-repair crash system:

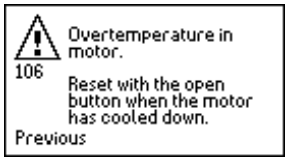
The door must be closed completely in hold to run mode. To do this the close button on the keypad must be pressed until the breakaway mechanism is snapped-in in place again and the door stops autonomous in the lower end position. Another “Open” impulse will then make the door ready to use again.

5.12.2 Manually repairable crash system:

The breakaway mechanism has to be snapped-in in the side frame manually. After this the door must be moved to the lower end position to quit the error, and it is then ready for use again.

5.13 Thermal switch

To protect the drive motor three thermal switches are fitted in its windings. If the switches trip the motor has become too hot and the drive is switched off immediately. The following message will then appear:

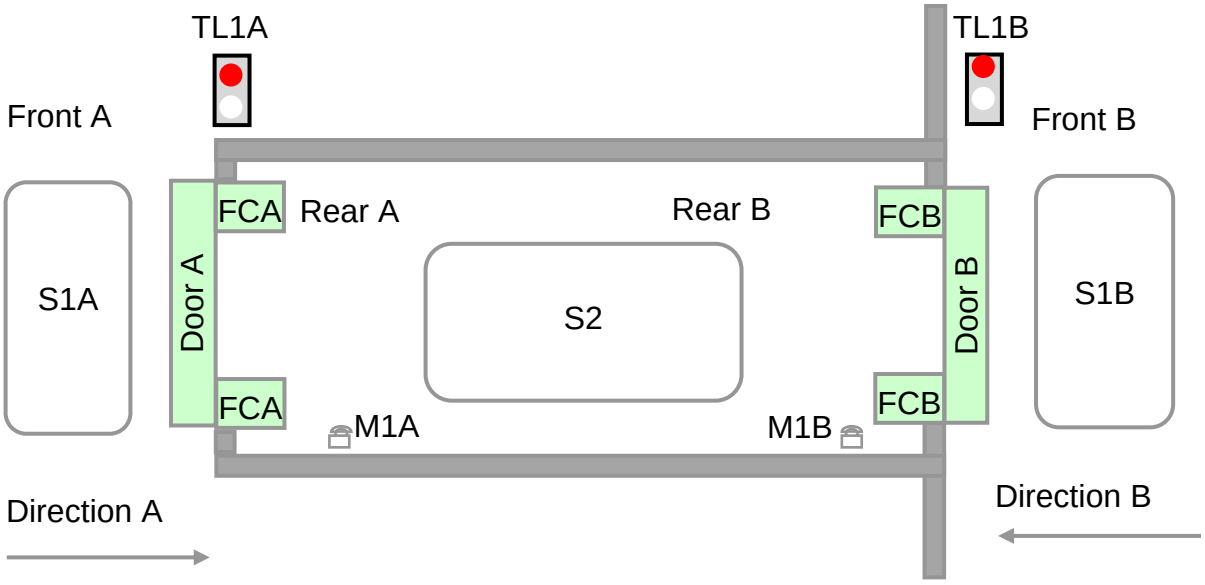


Once the motor has cooled down the error can be acknowledged by pressing the “Open” button. The door and the control system should then be checked for possible mechanical or electrical faults. The control system should not be switched back on until the motor has cooled down.

5.14 Airlock function [1]

An Airlock consists always of a door A and door B and is controlled automatically by the airlock sequence. Each door has a separate control system. An airlock is used, e.g. to prevent draughts in external doors.

5.14.1 Airlock sequence [1]



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#	**Description**
S1A	Actuator for Door A
S1B	Actuator for Door B
S2	Middle actuator in the airlock
FCA	Door line photocell Door A
FCB	Door line photocell Door B
M1A	Actuator for exiting the airlock Door A
M1B	Actuator for exiting the airlock Door B
TL1A	Traffic light for Door A
TL1B	Traffic light for Door B

5.14.2 Airlock passage from A to B with middle actuator

The airlock sequence can only be started if both doors are closed and the actuators are not activated. If the doors are equipped with traffic lights, both traffic lights must be red or off (depending on the parameter setting).

As soon as a vehicle activates actuator S1A, which is programmed with the function “Open with automatic closing after time (front)“, Door A opens. After the door has been opened the airlock can be driven into. The door closes automatically after driving into the airlock and after the close after passage time has expired or after the hold open time, if the airlock has not been driven into.

Door B automatically opens if Door A is closed and middle actuator S2 of the airlock has been actuated. The middle actuator pulse is saved with the airlock function. This means that door B opens, even if the middle actuator is no longer activated, before Door A is closed. The airlock can now be exited through Door B which automatically closes after the hold open time or the close after passage time (depending on the actuator). If another vehicle drives through Door B into the airlock before Door B is closed, the airlock can be driven through in the other direction (but middle actuator (S2) must be activated once). Direction detection in the airlock and therefore the door initiated by the activation of the middle actuator is controlled by the door line photocell. If the door line photocell is actuated and then the middle actuator is activated, the direction is detected.

5.14.3 Airlock passage from A to B without middle actuator

The airlock sequence can only be started if both doors are closed and the actuators are not activated. If the doors are equipped with traffic lights, both traffic lights must be red or off (depending on the parameter setting).

If a vehicle activates actuator S1A, which is programmed with the function “Open with automatic closing after time (front)“, Door A opens. After the door has been opened the airlock can be driven into. The door closes automatically after driving into the airlock and after the close after passage time has expired or after the hold open time, if the airlock has not been driven into.

Door B opens automatically if Door A is closed, regardless of whether the airlock has been entered or not. The airlock can now be exited through Door B which automatically closes after the hold open time or the close after passage time (depending on the actuator). If another enters the airlock through Door B before Door B has started to close, direction detection is not set and Door A does not open. Direction detection in the airlock is activated by the actuator which opened the closed door or opens it while it is closing. Here it is possible that somebody enters the

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airlock and the 2nd door does not open automatically. Therefore, in this airlock option, a manual actuator for exiting the airlock should be provided at each door within the airlock (M1A and M1B).

5.14.4 Long vehicle function at doors with middle actuator



This function is required if the vehicle to pass through the airlock is longer than the distance between the two airlock doors.

The first door opens and the long vehicle can drive into the airlock. If the door line photocell of the opened door and the middle actuator (S2) are simultaneously activated, the second door opens automatically and the airlock can be completely driven through. The first door closes after the close after passage time and the second door after the hold open or close after passage time (depending on the actuator). This function is also possible from the opposite side.

5.14.5 Manual long vehicle function

There is a possibility to have a separate long vehicle function, like a push button or pull rope, connected to door A , independent of a middle actuator is used or not. See Programming the inputs (chapter 4.14).

This long vehicle function is independent of the middle actuator and opens both doors without delay.

5.14.6 Permanent opening function

In several cases, e.g. during hot weather, it can be advantageous to use Door A only or Door B only. In this case the other door (here Door A) is set to permanent opening using the permanent opening function at the display (chapter 5.10) or, if installed, using the external permanent opening switch.

The opening pulse from the middle actuator (S2) then serves as a normal opening pulse for the door without permanent opening (Door B). The external actuator (here S1B) opens the door accordingly from the other side. The external actuator (here S1A) which is connected directly to Door A operated in permanent opening, does not have any function. In airlocks without middle actuator the external actuators (S1A or S1B) acts as opening pulse for the door without permanent opening.

5.14.7 Direction detecting red traffic light on the outside of the airlock door

The red traffic lights are connected depending on the function “Traffic light red 24 V DC front” (24 V DC traffic light) or “Traffic light (red/green) front” (traffic light via relay contacts). The traffic lights via relay contacts require an additional MCC output to switch off the traffic light if the door is closed.

Traffic light function settings: “Off if door is closed” or “Permanently on”

Airlock function from A to B:

Position of Door A	Traffic light function of Door A	Position of Door B	Traffic light function of Door B
Door A closed	Red => Off/On (*1) Green => Off	Door B closed	Red => Off/On (*1) Green => Off
Door A when opening	Red => On Green => Off	Door B closed	Red => On Green =>Off
Door A open	Red => Off Green => On	Door B closed	Red => On Green => Off
Door A when closing	Red => On Green => Off	Door B closed	Red => On Green => Off
Door A closed	Red => On Green =>Off	Door B when opening	Red =>On Green =>Off
Door A closed	Red =>On Green =>Off	Door B open	Red =>On Green =>Off
Door A closed	Red =>On Green =>Off	Door B when closing	Red =>On Green =>Off
Door A closed	Red=>Off/On (*1) Green =>Off	Door B closed	Red=>Off/On (*1) Green =>Off

(*1) => Depending on the setting selected for the traffic light.

5.14.8 Actuators and time setting

The minimum close after passage time, which is possible for a airlock door, depends on whether actuators S1A and S1B are detecting direction or not. Direction detecting actuators are, e.g. radar detectors and loops detectors. Non-direction detecting actuators are, e.g. infrared motion detectors and photocells.

After the airlock has been passed through (from A to B), actuator S1B sends a renewed opening pulse to door B if the close after passage time is set shorter than the time required to leave the actuator sensing range of S1B. This opening pulse starts the close after passage and hold open time again and in airlocks without middle actuator, Door A opens again after Door B is closed, if direction detection has been set with the opening pulse. Direction detecting actuators should be used to prevent this. Alternatively, the close after passage time can be set for so long that the object has the possibility of leaving actuator sensing range before the close after passage time has expired.

Due to saving of the middle actuator pulse even, e.g. a simple photocell can be used as the middle actuator. Direction detection is then set by the door line photocell as the middle actuator is not activated simultaneously.

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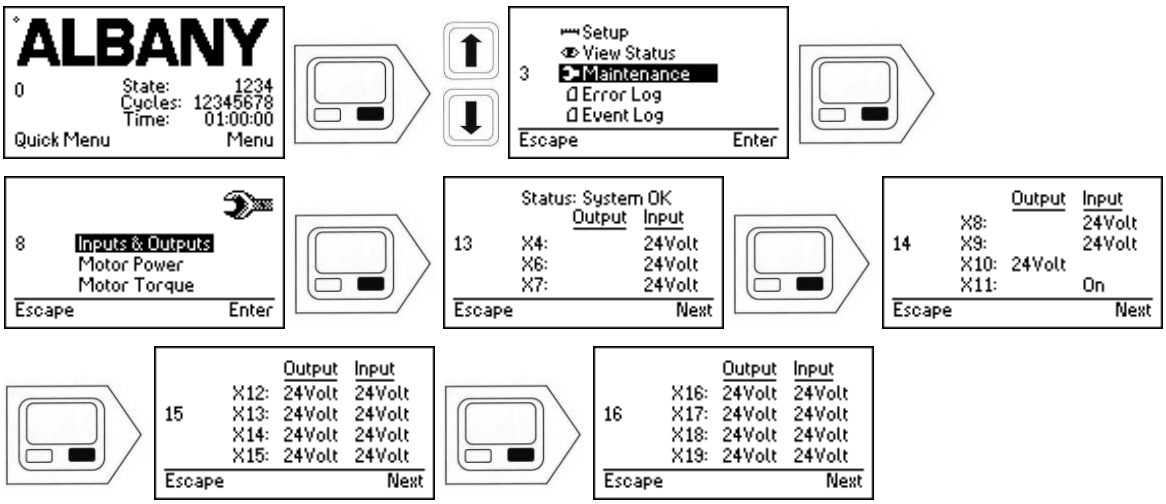
6 Fault correction

Contact the supplier or manufacturer for fault correction and repairs.

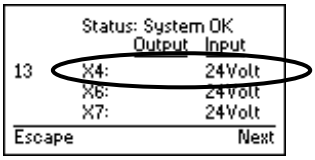
ASSA ABLOY Entrance Systems
Albany Door Systems GmbH
Service Department
Am Mondschein 25
59557 Lippstadt, Germany
Tel.: +49 (0)2941 766 0
Fax: +49 (0)2941 766 294
Internet: www.albanydoors.com
Email: service.de.albany@assaabloy.com

6.1 Checking the inputs and outputs

On these display screens it is easy to see what voltage is present at each input and output.



Example: In the electrical drawing the thermal switch is connected to X4. The following view appears on the display:

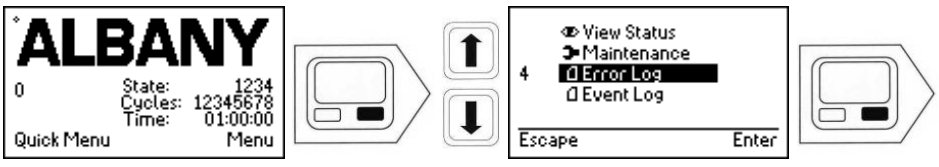


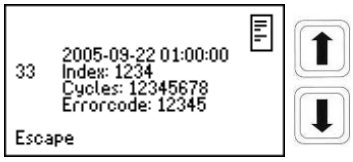
In this case the system status is OK, there is 24V at the input of the control system and the thermal contact appears to be in order.

6.2 Error log readout

Information and error messages are recorded in the error log of the control system with the date and time.

The error log of the MCC can be read-out as follows:





Every individual display view shows the date and time at which the error occurred, as well as the index, cycle status and the error code. The index is a running counter, which increases by one with each new error. The last occurred error is shown first. The two arrow keys on the keypad can be used to scroll through all error messages.

Whenever the main switch is switched off a low voltage warning will be recorded in the error log!

The error log is divided into 2 arrays.

0 – 500 array of information (e.g. configuration changed)

500 - … error messages (e.g. emergency manual handle actuated)

Error code	Description of error
0	Text entry
1	Error log erased
2	System start
3	Programming PC connected
4	Programming PC disconnected
5	Setup of end positions started
10	Invalid data in the EEprom
16	Door run 1 cycle (not activated)
17	Door run 10 cycles (not activated)
18	Door run 100 cycles (not activated)
19	Door run 1000 cycles (factory setting)
20	Reference limit switch position tolerance >10 impulses
21	Reference limit switch position tolerance >20 impulses
22	Reference limit switch position tolerance >50 impulses
23	Reference limit switch position tolerance >100 impulses
24	Reference limit switch position tolerance >150 impulses
30	IGBT temperature > 60°C (140°F)
31	IGBT temperature > 70°C (158°F)
32	IGBT temperature > 80°C (176°F)
33	IGBT temperature > 90°C (194°F)
34	IGBT temperature > 100°C (212°F)
41	Endurance test started
42	Endurance test stopped
43	State changed externally

44	Invalid positions
100	DSP, electrical safety edge activation
101	DSP, 18 V absent
103	DSP overvoltage warning
104	Overvoltage at DSP
105	DSP over temperature warning IGBT
106	Normal function (no reference search necessary)
107	Reference search after switch-on
108	Set up end positions
109	Overload warning
110	Safety function activated for closing impulse
111	DSP error
112	Valid door position saved
500	Invalid config file
503	Software logic error
504	DSP communication error
505	Software routines disturbed
506	Date or time out of valid range
507	Date/time out of valid range revised
508	Door moved from closed position without motor power
509	RAM test error
510	ROM test error
511	Closing speed higher than allowed
520-526	Communication error DSP – M16
1004	Electrical safety edge / pre-running photocell circuit interrupted during opening
1005	Electrical safety edge or the pre-running photocell activated
1006	Thermal contact tripped
1009	Movement of the door was not recognized
1010	Encoder registers unauthorised movement
1011	Emergency manual lever operated
1013	Reference limit switch activated out of valid range
1014	Door locked by other door (airlock)
1015	Crash sensor has tripped
1016	Pass door open
1018	Safety 1 defect
1019	Door line photocell defect
1022	Safety 2 defect
1023	Internal frequency converter error
1024	Electrical safety edge / pre-running photocell interrupted in door open position

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1025	Internal error
1026	Reference limit switch not activated with opened door
1027	Error in safety edge input
1028	Error in emergency stop input
1029	Input voltage at emergency stop input too high
1030	Earth fault in the 24 V circuit (low voltage)
1031	Earth fault in the 24 V circuit (high voltage)
1032	Opening height not reached during cyclical self-test (only for RR 300 Plus F+R)
1033	Breakaway function activated (only for US doors)
1100	Undervoltage – only in UPS mode activated
1101	Phase error
1102	Overvoltage
1104	Incorrect main voltage
1105	Overcurrent >25A
1106	Motor blocked or encoder without signal
1108	Over temperature
1110	Wrong hardware
1112	Brake circuit overcurrent
1115	3 Phase error
1200	Electrical safety edge or pre-running photocell activated

6.3 Event log readout

Important service messages and carried out service works are recorded in the control system's event log.

The event log of the MCC can be read-out as follows:



Every individual display view shows the date and time at which the event occurred, as well as the index, cycle status and the event code. The index is a running counter, which increases by one with each new event. The last occurred event is shown first. The two arrow keys on the keypad can be used to scroll through all event messages.

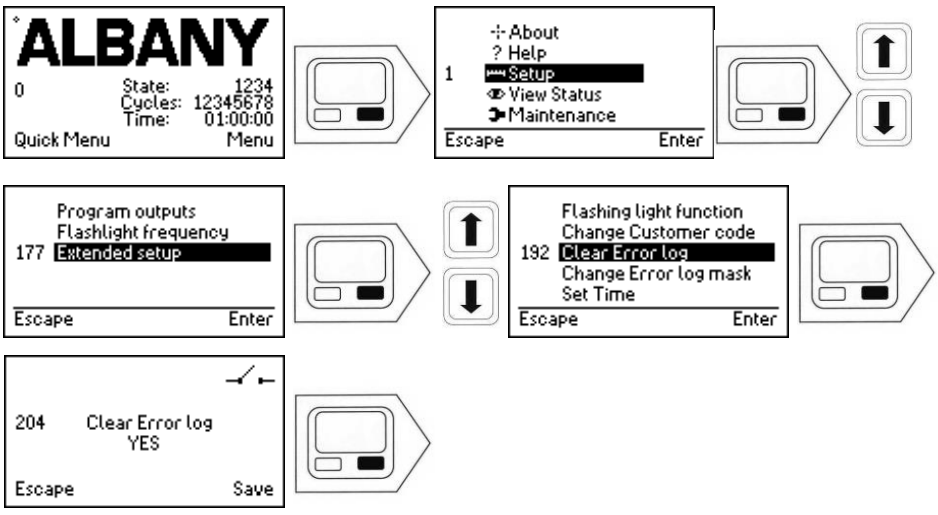
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The following actions are recorded in the control system's event log:

Event code	Event
1	Event log cleared
2	Config file changed
3	Application software changed
5	Information note in event log added or changed
10	Door service required
11	Door serviced
20	Control system cycle counter changed
21	Door cycle counter changed
22	Function of inputs/outputs changed
23	Time changed

6.4 Clear error log [3]

The error log should only be cleared by ASSA ABLOY Entrance Systems Service as all existing error messages are permanently erased.



6.5 Altering the error mask [3]

Only for ASSA ABLOY Entrance Systems Service Team.

6.6 Error messages / error codes (only for doors without display)

For doors without a display the most important error messages are shown by the flashing status light:

e.g. 3 flashes correspond to the error code position error.

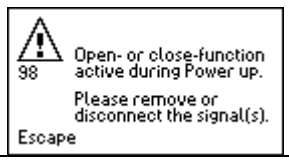
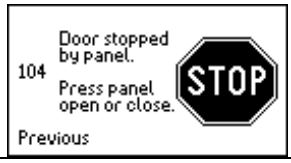
For purposes of clarity only the most important errors are signalized by codes of flashes by the status light (option).

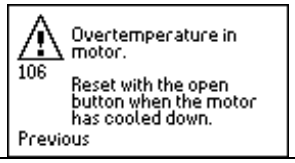
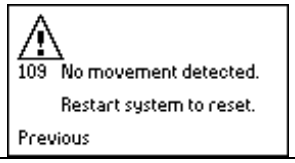
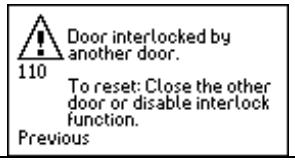
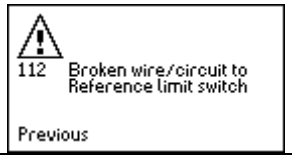
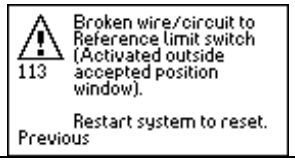
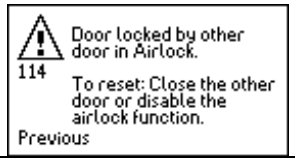
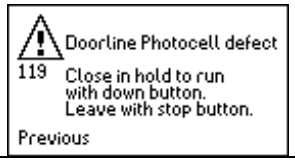
Error code	Error	Corrective action
1 flash (slow)	Stop button or emergency stop pressed, electrical safety edge activated according to the max consecutive activations setting in a row, interlock device activated, hold brake released	Press reset
2 flashes (slow)	Crash activated	For auto-repairing doors close the door, otherwise perform manual repairs
3 flashes (slow)	Position of door incorrect, lost positions after start-up	Perform reference search
4 flashes (slow)	Excess temperature of control system or motor, overload	Cool motor or control system
5 flashes (slow)	Sensor error	
6 flashes (slow)	The main voltage is set incorrectly.	Correct the main voltage setting in the installation menu
7 flashes (slow)	Internal error	Call the ASSA ABLOY Entrance Systems Service Team

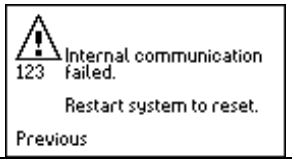
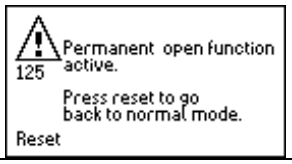
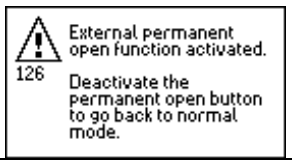
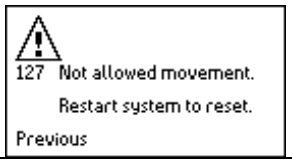
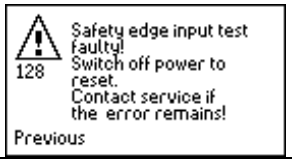
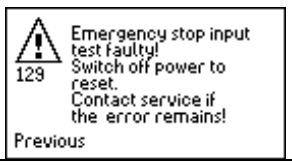
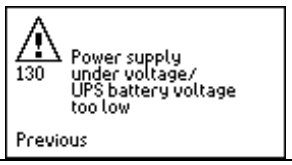
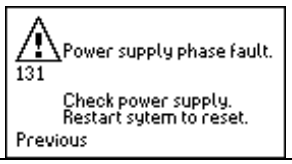
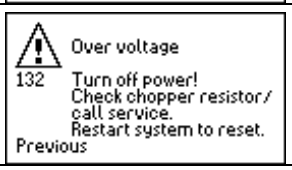
6.7 Error messages (only for doors with displays)

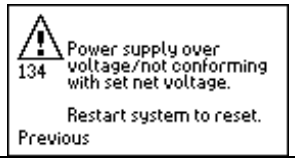
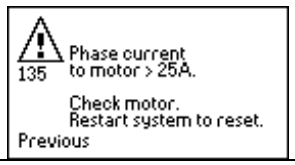
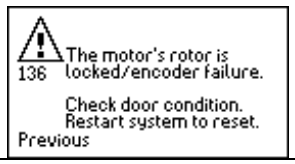
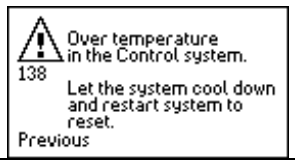
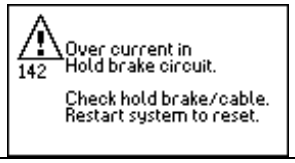
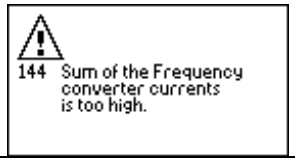
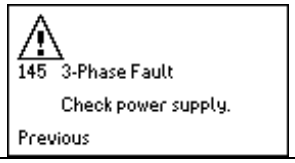
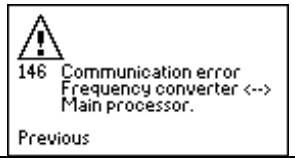
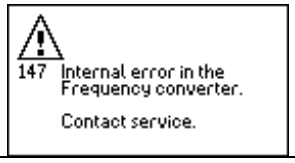
The following error messages are listed in ascending numerical order according to the display screen number. This number can be taken from the left-hand side of the display for every display message, regardless of the language. For example, 103 for “Door stopped“.

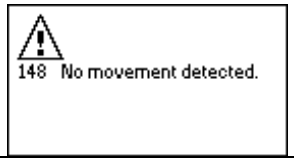
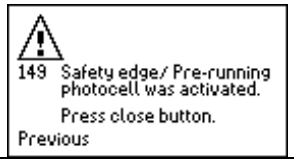
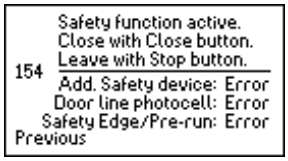
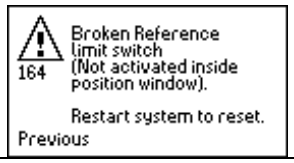
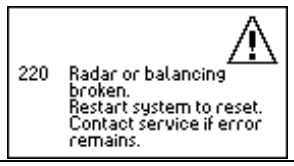
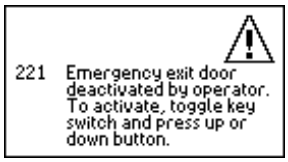
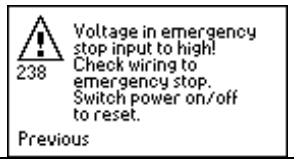
If no corrective action is specified, the error cannot be repaired or the error message appears again, please contact ASSA ABLOY Entrance Systems Service.

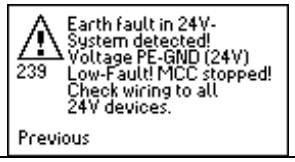
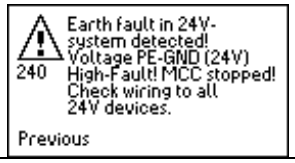
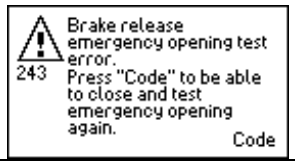
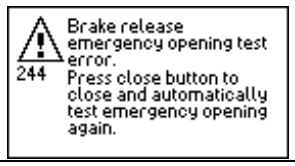
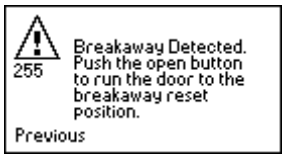
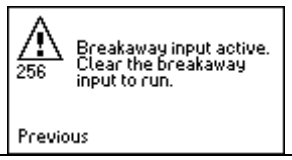
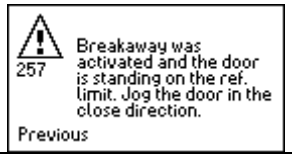
View-No	Display view	Corrective action
98		Check impulse actuator.
103		Release stop button if necessary. Acknowledge by actuating open or close command.
104		Acknowledge by actuating open or close button on the display.

105		Check pre-running photocell or electrical safety edge. Clean pre-running photocell. Acknowledge by actuating close button on the display.
106		Switch off the control system, allow the motor to cool and, if necessary, clean the motor's ventilation slots.
107		Internal safety test faulty. RAM/ROM tests, speed monitoring or other internal safety tests faulty.
109		Encoder gives incorrect signals. The motor or the top roll is blocked.
110		The door is locked by another door. Close the other door to reset this.
111		Return the emergency manual lever to its original position or remove the emergency hand crank.
112		Check wiring reference limit switch.
113		The reference limit switch was activated outside accepted door position window which is given by the encoder or the reference limit switch circuit is broken.
114		The door has been locked in place by the other air-lock door. To reset, close the other door completely or deactivate the air-lock function.
115		For doors without an auto-repair crash system close the door completely after repairs to acknowledge the error message. For doors with an auto-repair crash system close the door until the bottom profile engages automatically again and the lower end position is reached.
119		Exclude influence from other infrared (IR) sensors and check alignment of the door line photocell.

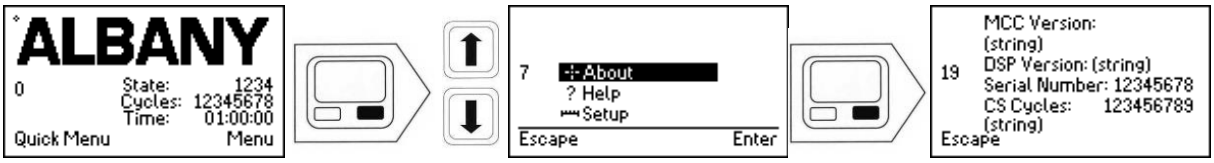
122		Check function and wiring of additional safety devices.
123		Please call the ASSA ABLOY Entrance Systems Service Team.
124		Check function and wiring of electrical safety edge / the pre-running photocell.
125		Permanent opening has been activated. Press left function key for reset.
126		External permanent opening has been activated. To deactivate this, remove the permanent opening signal.
127		The encoder detects an unauthorized movement of the door.
128		Please call the ASSA ABLOY Entrance Systems Service Team.
129		Please call the ASSA ABLOY Entrance Systems Service Team.
130		Undervoltage detected in the main voltage supply. Check the main voltage. Verify that the original config settings for the door are correct. Recharge the batteries in case of doors with UPS.
131		Check main voltage and supply stability. To reset switch the main switch off and back on again.
132		Overvoltage in the control system. Turn off the main switch and check the brake resistor, the main voltage and control settings of main voltage.

134		Check the supply voltage and the parameter for the main voltage.
135		The motor current is too high. Check that the door mechanism is running smoothly.
136		Check that the motor and the door mechanism is running smoothly. Check alignment and function of the encoder.
137		The control system is overloaded. Check that the door mechanism is running smoothly.
138		Overheating of the control system. Switch off the main switch and allow the control system to cool.
140		Please call the ASSA ABLOY Entrance Systems Service Team.
142		Check the brake and the brake cable.
144		Check that the door mechanism is running smoothly.
145		3-phase voltage failure or break-down of the supply voltage occurred. Check the supply voltage and the fuses.
146		Please call the ASSA ABLOY Entrance Systems Service Team.
147		Please call the ASSA ABLOY Entrance Systems Service Team.

148		The door is blocked or the position encoder is sending incorrect signals. Check that the door mechanism is running smoothly. Check the function of the encoder.
149		Check electrical safety edge / the pre-running photocell. Clean pre-running photocell. Press the “Open” or “Close” buttons to reset.
151		Release the emergency stop or remove emergency hand crank and start the door using the “Open” or “Close” command.
153		Acknowledge by actuating open command.
154		The door’s safety function is active. This means that there is something wrong with one or more of the door’s safety devices. The current status of the safety devices connected can be seen on the display. Once all safety devices are functioning correctly again you can exit or leave the mode by pressing the stop button.
164		Check reference limit switch and adjust if necessary.
220		Only for emergency exit door Albany RR 300 plus F+R Check springs, belts and spring/belt break switches. Check wiring of the radar and spring/belt break switches.
221		Only for emergency exit door Albany RR 300 plus F+R Toggle deactivation key-switch to switch-on emergency exit function. If message continues to be displayed, check the switch and wiring and replace if necessary.
222		Only for emergency exit door Albany RR 300 plus F+R No measures are necessary after successful completion. If performance of the self-opening-test is faulty, consider other fault messages.
238		The voltage at the emergency stop input is too high. Check wiring of emergency stop circuit.

239		An earth fault has been detected in the 24 V circuit. Check the wiring of all 24 V loads.
240		An earth fault has been detected in the 24 V circuit. Check the wiring of all 24 V loads.
243		Only for emergency exit door Albany RR 300 plus F+R Door can be released with right function key “Code” and input of the customer code. The self-tests are then repeated. The message 222 is indicated afterwards.
244		Only for emergency exit door Albany RR 300 plus F+R Press Close button for renewed checking of the self-opening height. No measures are necessary after successful completion.
255		Only for US doors: There has been a door impact. Acknowledge by moving door to upper end position of the door (breakaway position).
256		Check breakaway-sensors.
257		Follow instructions in the display view. Check the function and wiring of the reference limit switch.
260		Replace the batteries of the radio system only with equivalent batteries. Operation of the door in hold to run mode is possible.

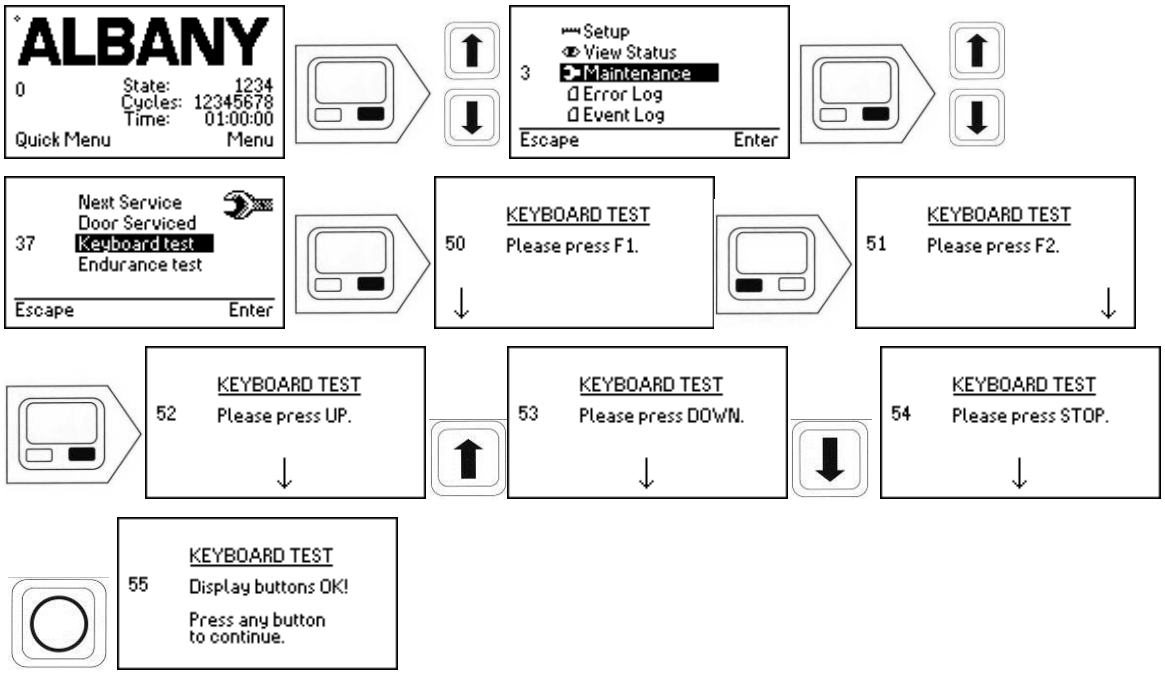
6.8 Hardware and software version readout



The hardware and software versions used can now be taken from the display.

6.9 Keypad test

This test is used to check whether the individual keys on the keypad are working.



6.10 Technical support

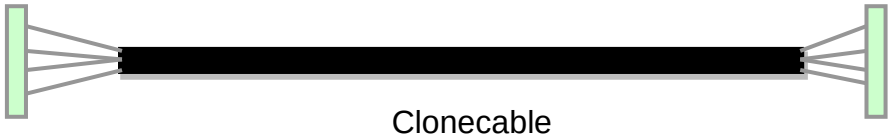


The display lists the contact information for technical support or the website on which the latest contact information for local technical support can be retrieved.

6.11 Clone function when replacing the control system

The software can be copied from one control to the other with the clone function. To clone a faulty MCC system the following components are needed:

- A new MCC control system with at least pre-installed boot software.
- The clone cable (therefore contact ASSA ABLOY Entrance Systems Service).
- An MCC display with cabling (if not already installed at the door)

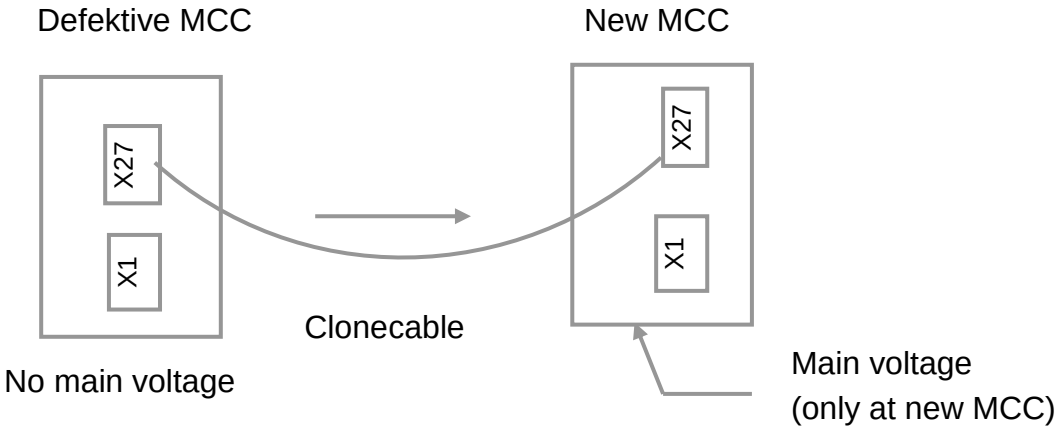


Clonecable

X27
(MCC-1)

X27
(MCC-2)

Status Approved 2018-04-17



At older control systems without the “Ready for Cloning” label, the changed customer settings in the display should be checked after each cloning procedure! These are, e.g. function changes at the inputs and outputs, etc.

Ready for Cloning √

“Ready for cloning” label

Follow the step-by-step instruction to perform a complete cloning:

The only cable that shall be connected to the faulty MCC is the cloning cable.

1. Turn off all power.
2. Disconnect and remove the broken MCC from the door.
3. Install the new MCC including the compl. electrical-connections.
4. Connect the cloning cable to both, the new MCC and to the faulty MCC (plug X27).
5. If no MCC display is normally connected to the door then temporary connect an extra one to the new MCC.
6. Turn on the power.
7. Select “CLONE” on the display.

   Wait until cloning is finished and OK is displayed. If the cloning has not been performed successfully the procedure can be restarted from point 6 by cycling the power.
8. Turn off the power.
9. Remove the faulty system, the cloning cable and eventually the extra MCC display.
10. Turn on the power.
11. Test the door.

7 Technical data and control system label

7.1 General data

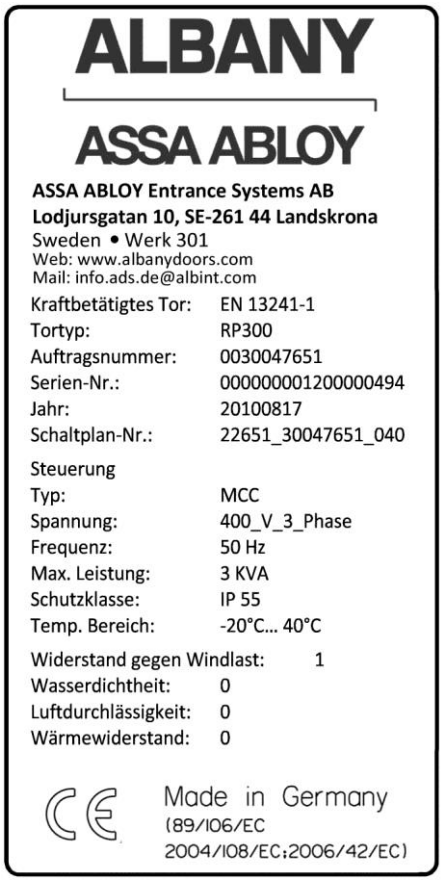
Housing material	anodized aluminium
Dimensions of housing WxHxD, without cable glands and end plates	220 x 380 x 100 mm (14,5" x 8,6" x 4")
Main voltage	3~ PE; 50-60 Hz 220-480 V -10%/+5%
Max. output power	5,5 kVA 2,3 kVA continuous average max. power
Max. output nominal current	8 A (FLA: Full Load Ampere) 12,7 A Boost for 0,5 sec.
Operating temperature range	-20°C - +40°C -20°C - +45°C RP-doors
Protection class	IP 55
External fuse by customer	10 – 16 Ampere B or C characteristic or comparable
Residual current protection (RCD`s) (only if needed)	AC/DC sensitive short-time delayed type B Observe the instructions in chapter 4.2.1!
Residual current caused by MCC	≤ 7mA according EN 60335-1
Control voltage / external supply	24 VDC / max. 500 mA with display max. 450 mA
Brake voltage	96 – 196 VDC / max. 300 mA
Programmable inputs	8
Programmable outputs	8
Weight	approx.. 5,2 kg
Conformity Europe	2006/42/EC, 2004/108/EC 2006/95/EC, EN13241-1: 2004 EN ISO 13849-1:2008 EN 12978:2009 EN 12453:2000 EN 60335-1:2002 TÜV Nord certification
Conformity USA and Canada	UL508C:2002 CSA22.2 No.247:1992
	File-Nr.: E326871
	Category: c UL us LISTED NMMS, NMMS2 c TÜV us NMMS7, NMMS8

Motor cable length	Without EMC filter: Max. 2m (industry and residential / commercial environments) With EMC filter: Max. 30m (industry and residential / commercial environments)

7.2 Classifications of safety circuits to EN ISO 13849-1

Function	Category	MTTFD	DCAVG	Performance Level
Input emergency stop	3	4633	90	D
Input electrical safety edge or the pre-running photocell	2	746	90	D

7.3 Labeling

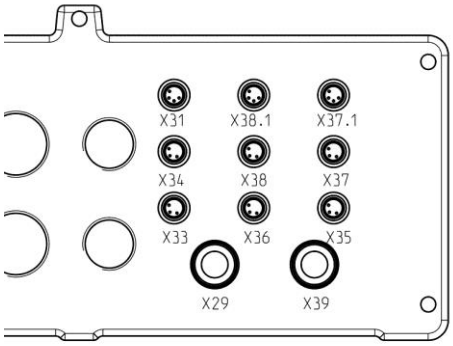


8 Optional MCC – Units

8.1 Flange plate with integrated junction box

The flange plate with integrated junction box allows an easy and time-saving wiring of the door sensors to the MCC .

8.1.1 Connector assignment junction box



If a safety edge is connected, it may be necessary to use a cable adapter (see electrical drawing)

DANGER

Serious injuries at the unsecured door. Loss of the CE conformity.

The inputs and outputs of the junction box and the MCC are interconnected parallel. A simultaneous use can lead to a loss in safety and to unexpected behaviour of the door.

- ⇨ The corresponding in- and outputs of the junction box and the MCC should not be wired/used parallel.
- ⇨ Observe electrical drawing of the door.
- ⇨ Simultaneous use according to order-related electrical drawing is allowed.

Terminal	Description
X 29	Encoder
X 31	Thermal contact/hand lever
X 33	Reference limit switch
X 34	Reserve I/O 5
X 35	Receiver, door line photocell
X 36	Transmitter, door line photocell
X 39	Display
At doors without crash:	
X 37	Receiver pre-running photocell or safety edge
X 38	Transmitter pre-running photocell
At doors with crash:	
X 37.1	Crash and receiver pre-running photocell
X 38.1	Crash and transmitter pre-running photocell

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8.2 Interface board

The MCC can be supplemented with a multifunctional interface board which increases the scope of the control system. There are 3 dry relay changeover contacts (8A, 250V~ / 10A, 240V-) on the interface board for controlling external components, which are controlled directly through the MCC outputs. In addition, the board offers the possibility to move 3 inputs from the MCC onto the interface board. This enables add-ons, e.g. loop detector or radio receiver to be mounted on the top-hat rail and to be directly wired with the inputs of the MCC. The terminal block X108 (terminal 1-14) can be used e.g. as an interconnection terminal block.

8.2.1 Terminal assignment of the interface board

DANGER



Risk of electrical shock

Infringement of protection class

⇨ If used, the protective earth terminal strip (earth) must be connected to the existing protective earth connection.



Cross-section, spring-loaded terminals X100-X103, X108, protective earth connection:

0.14-1.5 mm^2 without wire end sleeve,

0.14-1.5 mm^2 with wire end sleeve without plastic collar to DIN 46228-1

0.14-0.75 mm^2 with wire end sleeve with plastic collar to DIN 46228-4

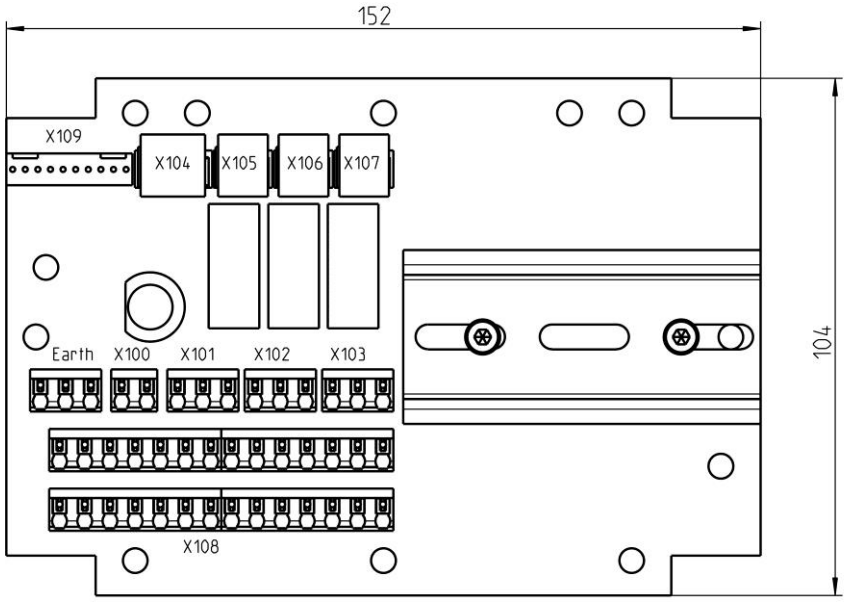


Cross-section, spring-loaded terminals X104-107:

0.14-1.5 mm^2 without wire end sleeve,

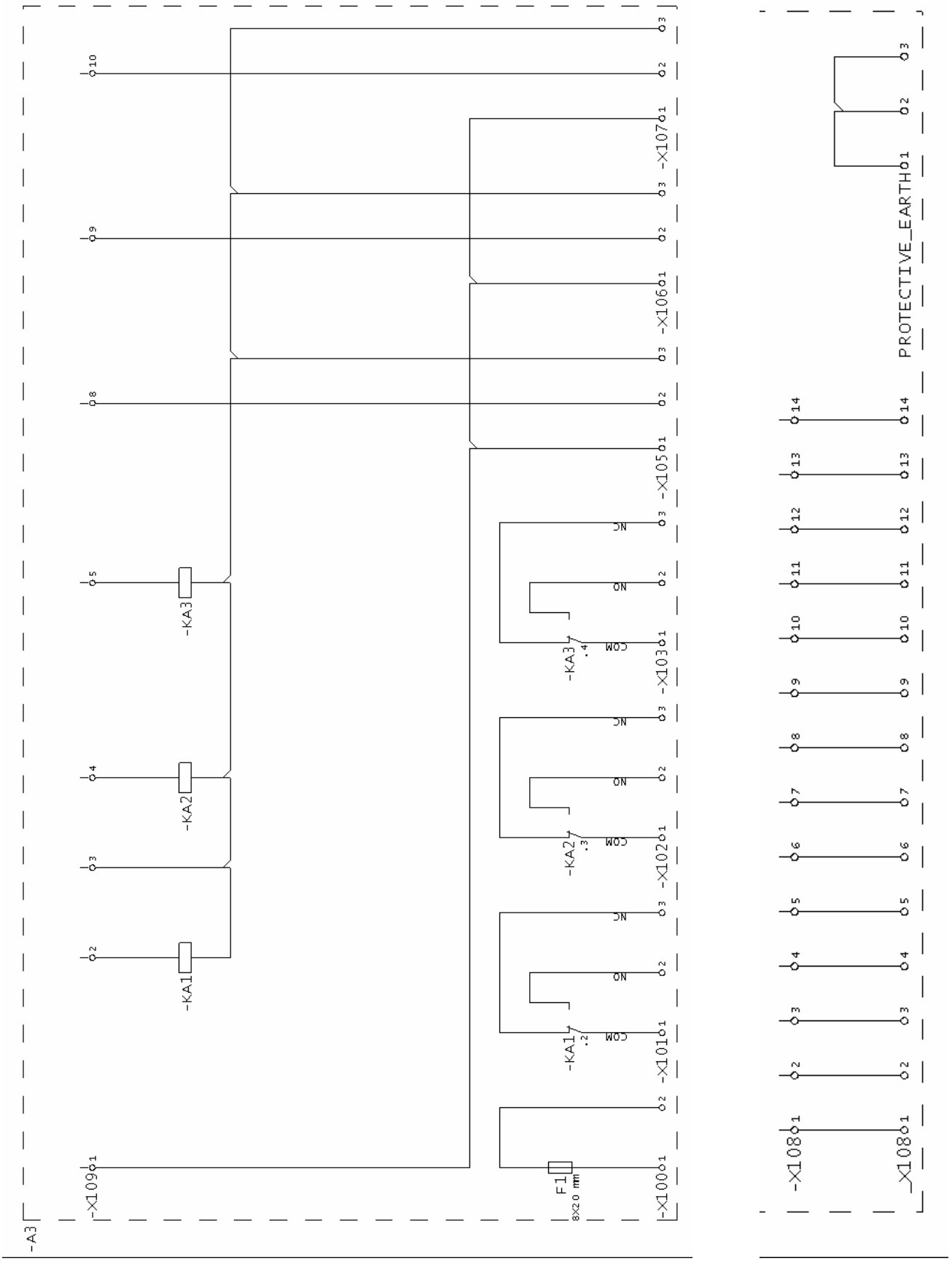
0.14-1.0 mm^2 with wire end sleeve without/with plastic collar to DIN 46228-1/-4

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Terminal	Description	Pin assignment
X 100	Connection of fuse F1	T2A => Terminal 1 and 2
X 101	Dry changeover contact of relay KA1	1=COM, 2=NO, 3=NC
X 102	Dry changeover contact of relay KA2	1=COM, 2=NO, 3=NC
X 103	Dry changeover contact of relay KA3	1=COM, 2=NO, 3=NC
X 104	(not used)	
X105-X107	Programmable inputs of the MCC	1= 24V DC, 2=Input, 3= GND, see order-related electrical drawing
X 108	interconnection terminal block with 14 terminals	
X 109	Connecting cable to the MCC	
Earth	Terminal for protective earth connection	

8.2.2 Schematic electrical drawing of the interface board:



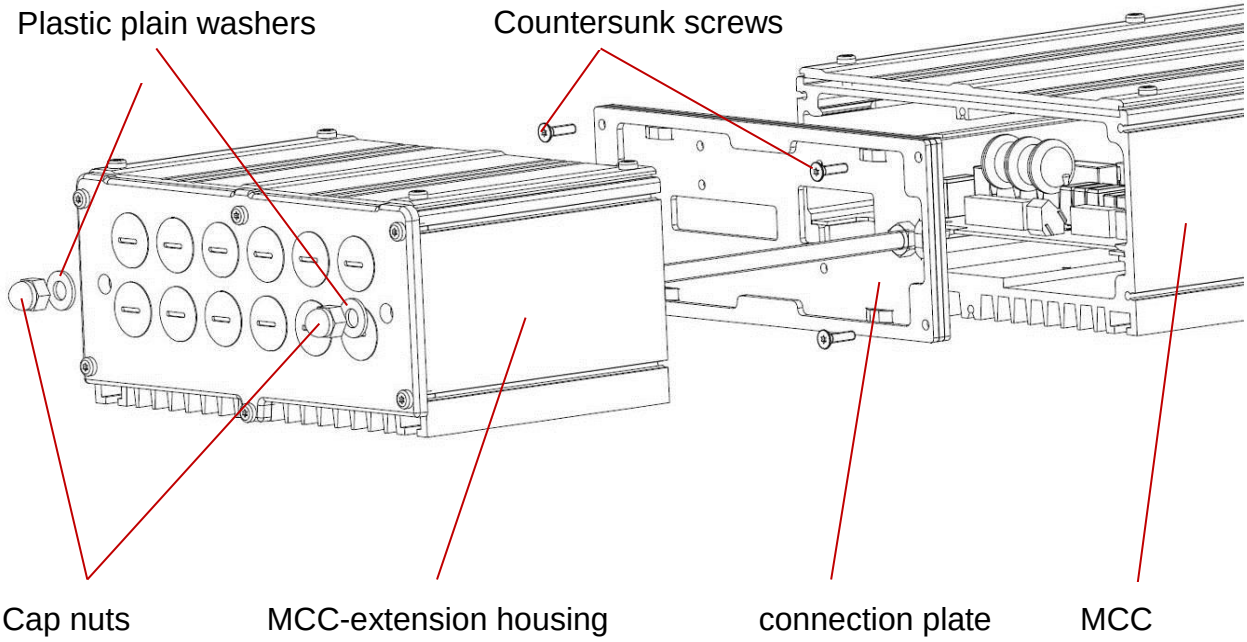
8.3 MCC extension

The MCC can be supplemented with a multifunctional extension which increases the scope of the control system and provides reserve space for additional components. The MCC extension enables various plug-on boards for radio receivers and loop detectors to be directly installed. The connection with the MCC is made using a 40-pin ribbon cable and therefore reduces the wiring required for the installation. With the MCC extension it is possible to connect the whole sensor/actuator wiring at the door to the extension and, at the customer's request, to guide the inlets for this through the extension housing.

There are 4 dry relay changeover contacts (6A, 250V~ / 3A, 30V-) on the extension board for controlling external components and which are controlled directly through the MCC outputs.

8.3.1 Mechanical installation of the MCC extension

As shown, the prepared connection plate must be screwed onto the MCC using the four countersunk screws. The extension housing is then pushed onto the connection plate and fixed using the two cap nuts.



CAUTION

Risk of material damage

Destruction of the MCC and the extension board by moisture.

⇨ To seal the housing, ensure that the two plastic plain washers are used under the two cap nuts

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8.3.2 Terminal assignment of the MCC extension board

DANGER

Serious injuries at the unsecured door. Loss of the CE conformity.

The inputs and outputs the extension board and the MCC are interconnected parallel. A simultaneous use can lead to a loss in safety and to unexpected behaviour of the door.

- ⇨ The corresponding in- and outputs of the extension board and the MCC should not be wired/used parallel.
- ⇨ Observe electrical drawing of the door.
- ⇨ Simultaneous use according to order-related electrical drawing is allowed.

DANGER

Risk of electrical shock

Infringement of protection class

- ⇨ If used, the protective earth terminal strip (earth) must be connected to the existing protective earth connection.

Cross-section, spring-loaded terminals X9-21, X216-217:

0.14-1.5 mm^2 without wire end sleeve,

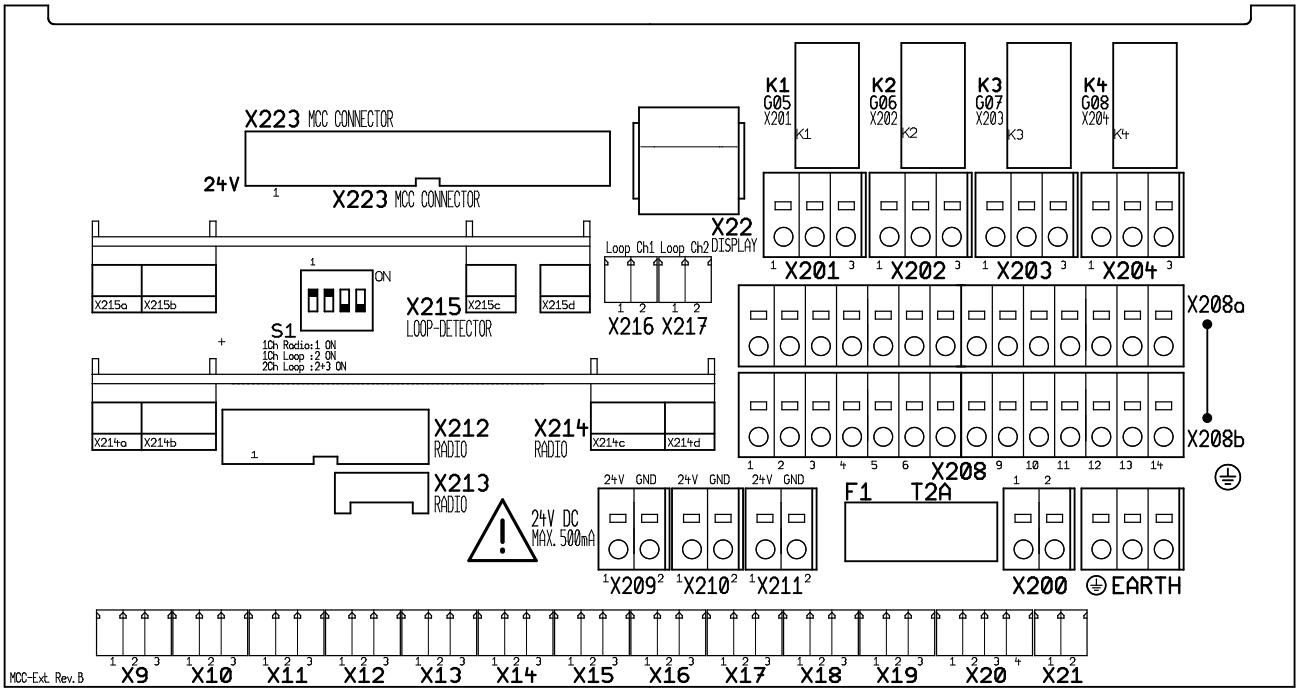
0.14-1.0 mm^2 with wire end sleeve without/with plastic collar to DIN 46228-1/-4

Cross-section, spring-loaded terminals X200-211, protective earth connection:

0.14-2.5 mm^2 without wire end sleeve,

0.14-2.5 mm^2 with wire end sleeve without plastic collar to DIN 46228-1

0.14-1.5 mm^2 with wire end sleeve with plastic collar to DIN 46228-4



Terminal	Description	Pin assignment
X 223	Plug-in connector for connection with the MCC	
X 9 – X 21	Cable connectors for sensors/actuators parallel to the plugs on the MCC	
X 22	Alternative connector for display	
X 200	Connection of the internal fuse F1 (250V T2A)	
X 201	Changeover contact relay K1 - MCC output GO5/X16	1=COM, 2=NO, 3=NC
X 202	Changeover contact relay K2 - MCC output GO6/X17	1=COM, 2=NO, 3=NC
X 203	Changeover contact relay K3 - MCC output GO7/X18	1=COM, 2=NO, 3=NC
X 204	Changeover contact relay K4 - MCC output GO8/X19	1=COM, 2=NO, 3=NC
X 208	interconnection terminal block with 14 terminals Terminals X208a and X208b are individually bridged internally!	
X209– X211	Supply terminals 24V DC for external loads	
X212 - X214	Slot for plug-in radio receiver	
X 215	Slot for loop detector 1 channel / 2 channel	
X 216	Terminal for loop 1 (twist loop wire up to the terminal!)	
X 217	Terminal for loop 2 (twist loop wire up to the terminal!)	
Earth	Terminal for protective earth connection	

8.3.3 Plug-on modules for MCC extension

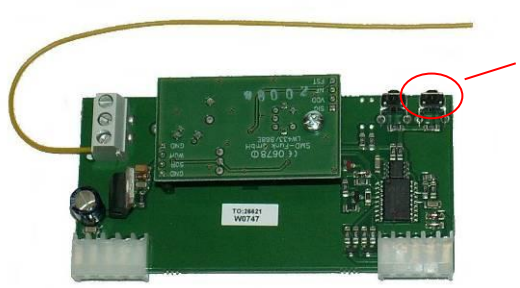
The MCC extension enables various plug-on modules for radio receivers and loop detectors to be directly installed. All slots provided for this are specially designed for the described plug-on modules. Plugging in other modules can cause damage or destruction of the MCC control system, the extension as well as the add-on modules. The plug-on modules may only be plugged in or pulled off if the control system is switched off (disconnected from the power supply).

The respective plug-on modules can only be attached to the corresponding slot and only in one direction on the extension. If it is necessary to remove a plug-on module from the socket, this is made easier by alternately tilting the module.

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8.3.3.1 Radio receiver 433 MHz

2 channel 433 MHz radio receiver, whereby only one channel is used on the MCC extension.



Teaching-in of the radio receiver:

Press teach in button of radio receiver and then actuate transmitter.

Up to 32 transmitter can be teached-in

To delete the teached-in transmitter press teach-in button until LED lights up!

Observe manual of the radio receiver!

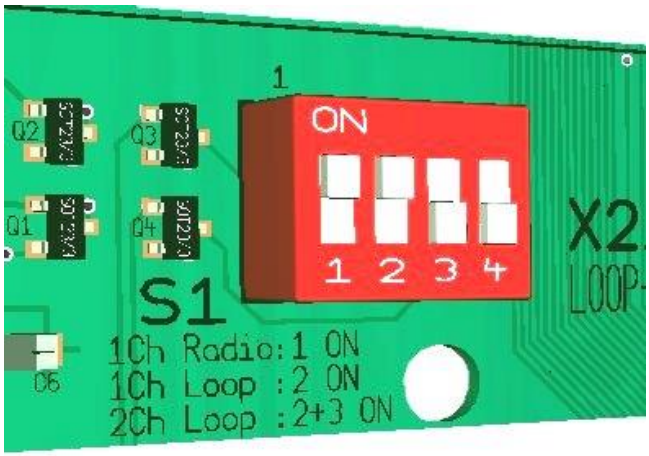
8.3.3.2 Loop detectors

Loop detector 1 channel / 2 channel as actuators for the detection of vehicles.



Observe the manual of the loop detectors!

8.3.3.3 Setting the slide switches for plug-on modules



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WARNING

Risk of personal injury and material damage. Disturbance of the general operation and unexpected behaviour of the door.

The inputs and outputs of the extension board and the MCC are interconnected parallel. A simultaneous use of the inputs and outputs can lead to a disturbance of the general operation and unexpected behaviour of the door.

- ⇨ The appropriate inputs to the MCC and the extension board should not be used for other impulse actuators when using the plug-on modules.
- ⇨ Observe electrical drawing of the door.
- ⇨ Double assignment according to order-related electrical drawing is allowed.

Radio receiver 1 channel

Switch slide switch 1 to ON position!

With the switching on of dip switch 1, input GI8 (X19) of the MCC is in use and can be used for other impulse actuators in exceptional cases only.

Loop detector 1 channel

Switch slide switch 2 to ON position!

With the switching on of dip switch 2, input gi7 (x18) of the MCC is assigned and can be used for other impulse actuators in exceptional cases only.

Loop detector 2 channel

Switch slide switches 2 and 3 to ON position!

With the switching on of dip switches 2 and 3, the inputs GI6 and GI7 (X17,X18) of the MCC are assigned and can be used for other impulse actuators in exceptional cases only.

Slide switch 4 does not have any function!

8.4 Uninterruptible power supply (UPS)

The UPS is a unit that gives the door control system power in case of a main power failure. The UPS is designed especially for MCC and is connected directly to the DC-link circuit to guarantee the efficient and reliable function of the UPS. The MCC controls the UPS.

In case of a main power failure the door can be operated several times immediately after the failure. Alternatively it is one time up to 4 hours with the UPS in idle mode. (Precondition: the batteries have to provide the specified capacity)

For efficiency reasons the door will operate with reduced speed in UPS mode of approx. 20Hz / 0,3m/s, depending on door type.

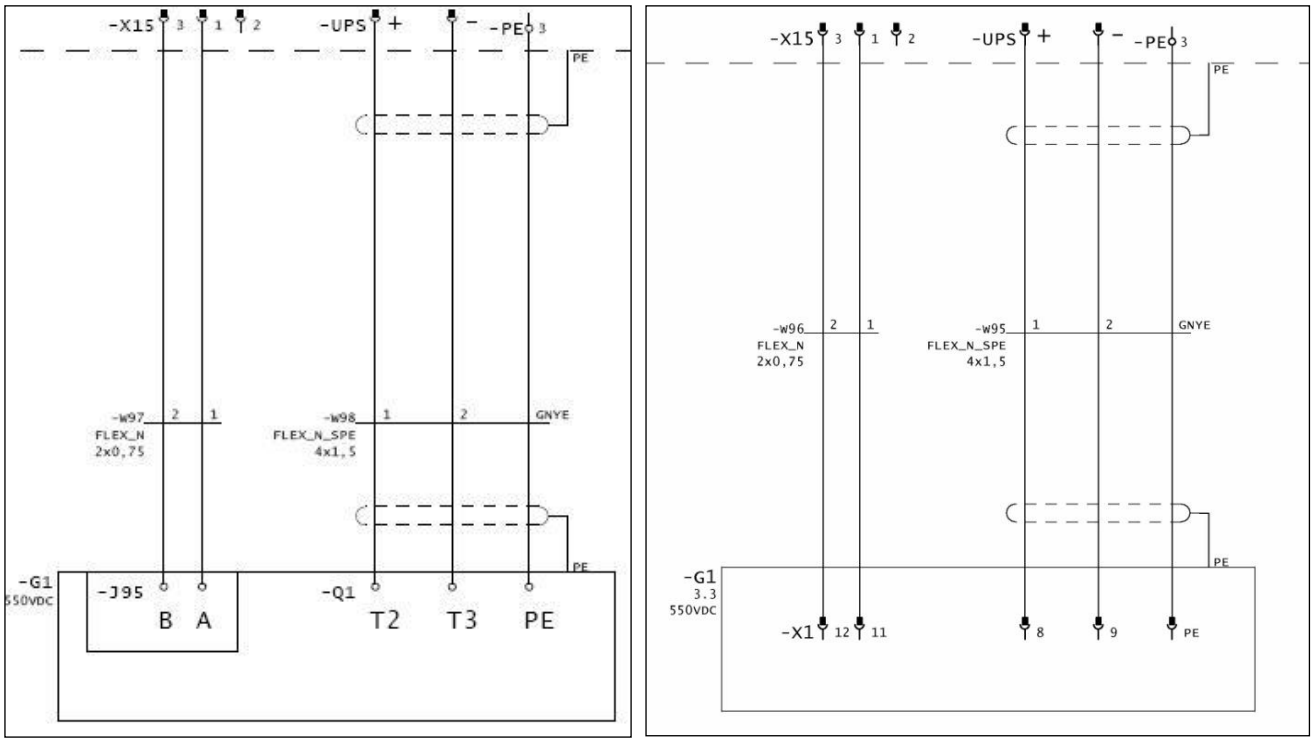
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8.4.1 Wiring the UPS to the MCC

The wiring of the door is done according to the electrical drawing. If there is no electrical drawing available, the connections can be made according to the following examples.

Programmable outputs X14-X19 of the MCC can be chosen to control the UPS. In the examples of the two versions the output X15 of the MCC controls the UPS .

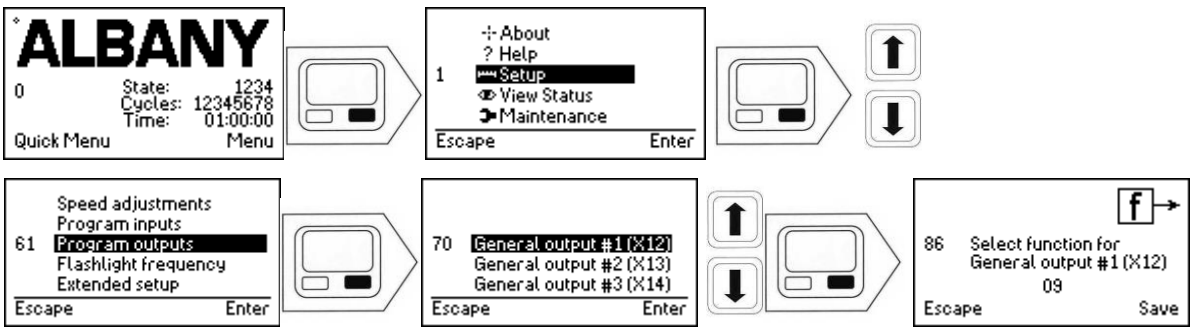
The UPS supply cable is to be connected to the two 6,3mm (1/4”) flat connectors marked with UPS+ and UPS- in the MCC.



8.4.2 Parameterisation of the MCC

Activating the UPS function requires two settings to be modified. The UPS control signal shall be programmed on a programmable output and the UPS function shall be set in the MCC.

8.4.2.1 Programming an output to give a control signal to the UPS



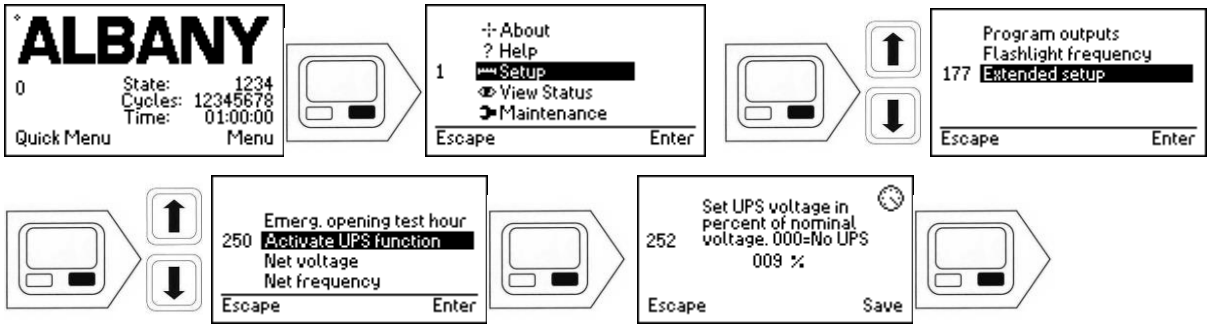
Select the output to be programmed (according electrical drawing) and enter function code 72.

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The programmable outputs 1 and 2 (X12 and X13) are reserved for safety devices and therefore cannot be used for controlling the UPS.

8.4.2.2 Configure the MCC for the UPS function

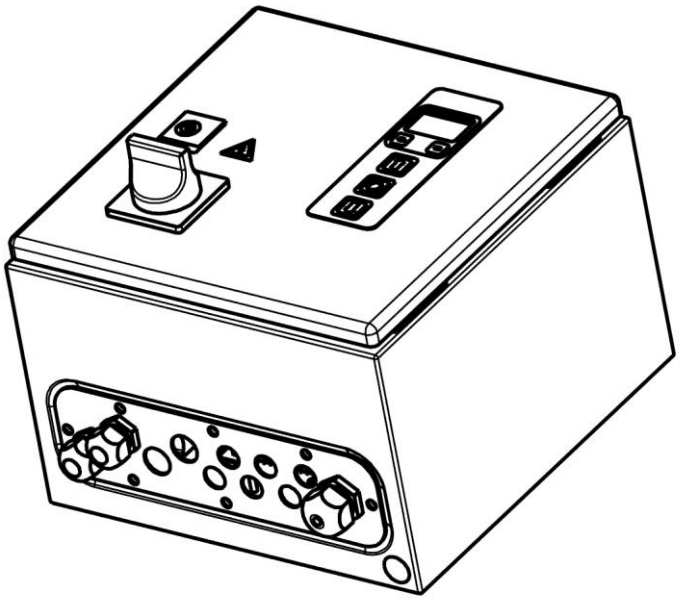
The UPS function needs to be activated in the MCC. It is done by adjusting the UPS voltage in relation to the nominal net voltage in percentage. When no UPS is connected is this value adjusted to 0%. Normal value with the UPS connected is 100%.



8.5 Single-Phase Combibox

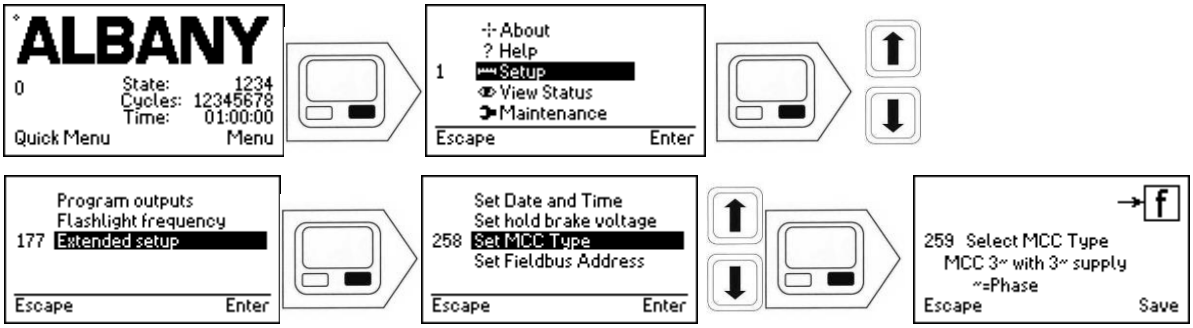
By the use of the “single phase” combibox the MCC can be operated at single-phase main supplies.

The operation of several doors at one phase should be avoided due to the high current load of the single-phase supply. The main supply should be accordingly dimensioned. Recommended is a cross-sectional area of min. 2.5 mm^2.



The wiring of the "single phase” combibox has to be done according to the electrical drawing.

8.5.1 Parameterisation of the MCC for single phase main supply

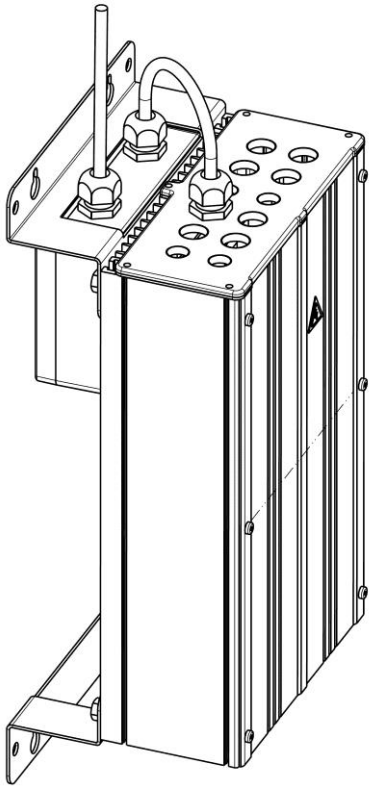


Select „MCC3~ with 1~ supply“ with Open/close button.

8.6 EMC filter

For motor cable length from 2m to 30m a EMC filter is necessary, in order to keep the EMC requirements for the industrial and residential / commercial environments.

The EMC filter is in a separate housing and mounted below the MCC. Electrical wiring is to be made between MCC and the main supply. Observe the electrical drawing of the door.



CAUTION

Loss of the CE conformity.

Without EMC filters the requirements of the European EMC directive are not kept and it may come to disturbances of other electrical devices.

⇨ EMC filter is required for motor cable length larger than 2 m.

8.7 Field bus module

The field bus module of the MCC offers the possibility to connect the MCC to different field bus systems. The corresponding fieldbus standard is set by selecting the used Anybus Compactcom (Anybus-CC) communication module.

8.7.1 Structure of field bus module

The fieldbus module consists of a mainboard, an Anybus CC communication module and the MCC extension housing, which is flanged at the MCC.

The communication of the field bus module with the MCC control system is done by a 10 pole flat cable. The 24 V supply is from the MCC and is to be connected to the plug X2 of the mainboard.

Depending on the field bus system the connection to the field bus is done by a M12 female or male plug which are installed in the flange-plate.

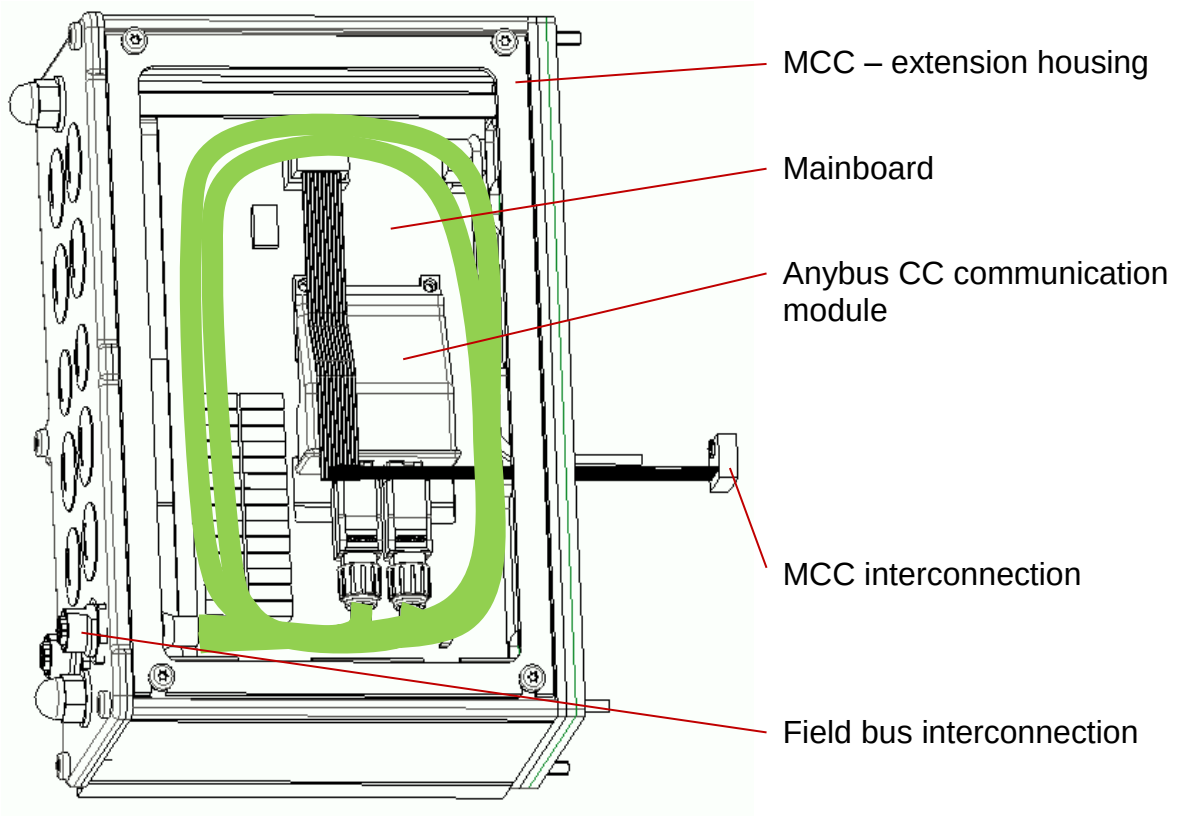
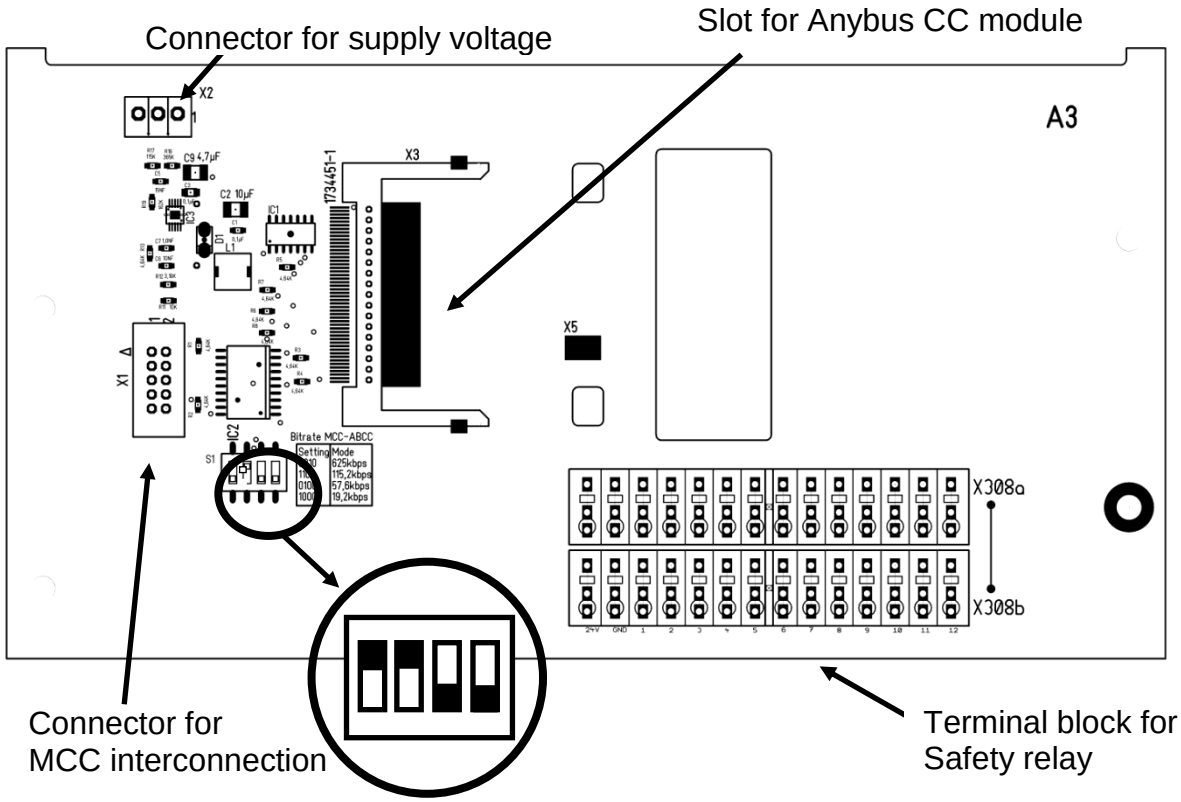


Figure 3: Field bus module with Profinet 2-Port

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8.7.1.1 Mainboard



The DIP-switches on the mainboard for the transfer rate between MCC and bus module should be always set to 115,2 kbps (1100).

8.7.1.2 Profinet 1-Port

For the Profinet 1-port bus structure an external network-switch is necessary.

For this solution the connection to the field bus is done via the flange plate with the integrated M12 female plugs (D-coded).

8.7.1.3 Profinet 2-Port

The network-switch for the Profinet 2-port bus structure is integrated in the bus module. With this solution the connection to the field bus is done via the flange plate with the two integrated M12 female plugs (D-coded).

8.7.1.4 Profibus

For the Profibus structure the connection to the field bus is done via the flange plate with the two integrated M12 female/male plugs (B-coded). The bus termination on both ends for the Profibus can be turned on and off with the switch in the 9-pole D-Sub connector.

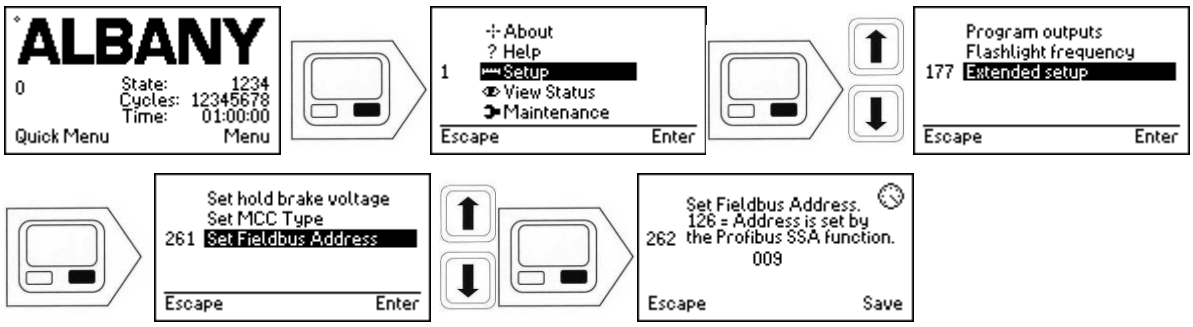
With the activation of the bus termination the outgoing cable is disconnected at the plug!

The male plug in the flange plate is for the incoming cable and the female plug for the outgoing cable.

A valid address range from 0 to 125 is possible. The address 126 is reserved for the SSA (Set Slave Address) function.

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8.7.2 Set Fieldbus Address



8.7.3 Generic Station Description (GSD-file)

The Generic Station Description (GSD-file) of the used bus module can be downloaded at http://www.anybus.com/products/abcc.shtml.

The user manual and the specifications of the manufacturer must be observed.

8.7.4 Ethernet/IP

The TCP/IP configuration setup can be made with the Anybus IP config Utility for module TCP/IP configuration. The utility as well as the EDS file can be downloaded from http://www.anybus.com/products/abcc.shtml.

8.7.4.1 Setup of the Ethernet/IP-Master

The settings shall be done according to the table below, which conforms to the information in the EDS file. These terms are also used in the RSLogix 5000.

	Assembly Instance	Size	Format
Input	100	2	16 bit
Output	150	2	16 bit
Configuration	5	0	8 bit

8.7.5 Bus communication

The cyclic write and read process data, in Profibus and Profinet called IO data will be mapped in to so call ADI's, application data instances. These are 16bit unsigned variables that can also be used for acyclic data when mapped so.

8.7.6 TX data from the MCC

8.7.6.1 Application Data Instance ADI[0]

Byte	2								1							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Bit position	**Rules**	**Value**	**Comment**
1: Door ready for work	0: Not ready 1: Ready	0x0001	
2:Door is closed	0: Door is not closed 1: Door is closed	0x0002	
3: Door is open	0: Door is not open 1: Door is open	0x0004	
4	For future use	0x0008	

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5	For future use	0x0010	
6	For future use	0x0020	
7	For future use	0x0040	
8	For future use	0x0080	
9: Custom 1		0x0100	Custom1-8: For customized digital data out from the MCC (see electrical drawing).
10: Custom 2		0x0200	
11: Custom 3		0x0400	
12: Custom 4		0x0800	
13: Custom 5		0x1000	
14: Custom 6		0x2000	
15: Custom 7		0x4000	
16: Custom 8		0x8000	

8.7.6.2 Application Data Instance ADI[1]

Byte	2								1							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

The ADI[1] data contains a 16 bit unsigned value of the current state of the MCC door controller. These data can be used for detecting normal operation states and for the error messages. A list of all the error states follows below.

8.7.6.3 Allocation of the MCC error messages

MCC error messages	**Decimal error state number**	**Hexadecimal error state number**
Electrical safety edge / pre-running photocell interrupted	1004	3EC
Electrical safety edge or the pre-running photocell activated	1005	3ED
Thermal contact tripped	1006	3EE
Encoder gives incorrect signals	1009	3F1
Encoder registers unauthorised movement	1010	3F2
Emergency lever/hand crank actuated	1011	3F3
Reference limit switch circuit interrupted	1013	3F5
Crash sensor has tripped	1015	3F7
Emergency stop actuated (X7)	1016	3F8
Safety 1 defect	1018	3FA
Door line photocell interrupted	1019	3FB
Safety 2 defect	1022	3FE
Internal frequency converter error	1023	3FF
Safety edge interrupted	1024	400
Internal error	1025	401
Reference limit switch interrupted	1026	402
Error in safety edge input	1027	403
Error in emergency stop input	1028	404
Input voltage at emergency stop input too high	1029	405

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Earth fault in the 24 V circuit (low voltage)	1030	406
Earth fault in the 24 V circuit (high voltage)	1031	407
Opening height not reached during cyclical self-test (only for RR300 Plus F+R)	1032	408
Breakaway function activated	1033	409
Frequency converter undervoltage	1100	44C
Frequency converter phase error	1101	44D
Frequency converter overvoltage	1102	44E
Frequency converter overvoltage warning	1103	44F
Frequency converter error, wrong main voltage	1104	450
Frequency converter overcurrent	1105	451
Frequency converter error, motor blocked	1106	452
Frequency converter error, IGBT overload	1108	454
Frequency converter error, wrong hardware	1110	456
Frequency converter error, brake circuit overcurrent	1112	458
3 phase error, frequency converter	1115	45B
Electrical safety edge or pre-running photocell activated	1200	4B0

8.7.6.4 Byte order Profibus and Profinet IO

Both are MSB (Most Significant Byte) first buses, so the byte order is:

Byte 1	Custom data from the MCC (chapter 8.7.6.1)
Byte 2	Standard data from the MCC (Door is open etc.) (chapter 8.7.6.1)
Byte 3	MSB of the current MCC state (chapter 8.7.6.2)
Byte 4	LSB (Least Significant Byte) of the current MCC state (chapter 8.7.6.2)

8.7.7 RX data, data to the MCC

8.7.7.1 Application Data Instance ADI[2]

Byte	2								1							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Bit position	**Rules**	**Value**	**Comment**	**Reset**
1: Open door (UP)	0: NOP 1: Open the door	0x0001		Lost bus communication or the Anybus module is in faulty
2:Stop door (Stop)	0: NOP 1: Stop the door	0x0002		Lost bus communication or the Anybus module is in faulty
3: Close the door (close)	0: NOP 1: Close the door	0x0004		Lost bus communication or the Anybus module is in faulty
4: Open the door to reduced	0: NOP 1: Open the door to	0x0008		Lost bus communication or the Anybus module is in faulty

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height	reduced height			
5 Reset system	0: NOP 1: The system will be reset	0x0010	This bit is used to reset the control from error states. This bit shall only be active for a maximum time of 500ms to avoid that the control is restarted several times.	Lost bus communication or the Anybus module is in faulty
6	For future use	0x0020		
7	For future use	0x0040		
8	For future use	0x0080		
9: Custom 1		0x0100	Custom1-8: For customized digital data in to the MCC (see electrical drawing)	Custom1-8: Lost bus communication or the Anybus module is in faulty
10: Custom 2		0x0200		
11: Custom 3		0x0400		
12: Custom 4		0x0800		
13: Custom 5		0x1000		
14: Custom 6		0x2000		
15: Custom 7		0x4000		
16: Custom 8		0x8000		

8.7.7.2 Application Data Instance ADI[3]

Byte	2								1							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Bit Pos.	**Rules**	**Value**	**Reset**
1	For future use	0x0001	Lost bus communication or the Anybus module is in faulty
2	For future use	0x0002	
3	For future use	0x0004	
4	For future use	0x0008	
5	For future use	0x0010	
6	For future use	0x0020	
7	For future use	0x0040	
8	For future use	0x0080	
9	For future use	0x0100	
10	For future use	0x0200	
11	For future use	0x0400	
12	For future use	0x0800	
13	For future use	0x1000	
14	For future use	0x2000	
15	For future use	0x4000	
16	For future use	0x8000	

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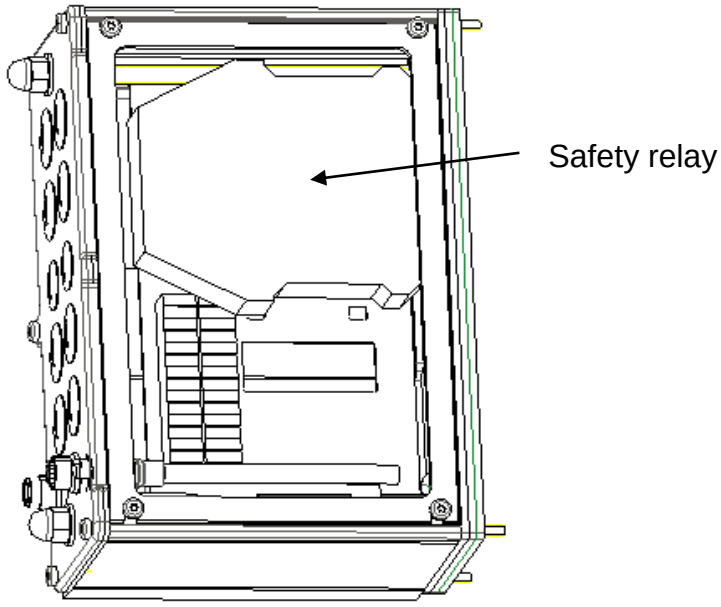
8.7.7.3 Byte order Profibus and Profinet IO

Both are MSB (Most Significant Byte) first buses, so the byte order is:

Byte 1	Custom data to the MCC (chapter 8.7.7.1)
Byte 2	Standard data to the MCC (Door open etc.) (chapter 8.7.7.1)
Byte 3	For future uses (chapter 8.7.7.2)
Byte 4	For future uses (chapter 8.7.7.2)

8.7.8 Installation of the safety relays (Option for machine protection doors)

Additional to the bus module it is possible to mount a safety relay for the safety limit switches at the machine protection door in the MCCC extension box. This relay is snapped on the DIN rail of the MCC adapter plate. The wiring is done via the terminal blocks X308a and b on the mainboard.



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